A

PROJECT REPORT

(PS04CAST55)

ON

“A Comprehensive Statistical Analysis and Price Prediction of Airline Ticket Pricing Factors”

A

PROJECT REPORT

SUBMITTED IN PARTIAL

FULFILMENT OF THE REQUIREMENTS

FOR THE AWARD OF THE DEGREE OF

M.Sc. in Applied Statistics

SUBMITTED BY

BOTADARA VISHAL BHARATBHAI

(ROLL NO:18)

UNDER THE GUIDANCE OF

Ms. MANSI BALVA

DEPARTMENT OF STATISTICS

SARDAR PATEL UNIVERSITY

VALLABH VIDYANAGER, ANAND – 388120

GUJARAT (INDIA)

April-2026

A COMPREHENSIVE STATISTICAL ANALYSIS AND PRICE PREDICTION OF AIRLINE TICKET PRICING FACTORS

A
PROJECT REPORT
SUBMITTED IN PARTIAL
FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF

MASTER OF SCIENCE

In

APPLIED STATISTICS

By

BOTADARA VISHAL BHARATBHAI
(Roll No.:18)

UNDER THE GUIDANCE OF

Ms. Mansi Balva

DEPARTMENT OF STATISTICS
SARDAR PATEL UNIVERSITY
VALLABH VIDYANAGAR - 388120
GUJARAT (INDIA)

APRIL-2026

A

PROJECT REPORT

(PS04CAST55)

ON

"A Comprehensive Statistical Analysis and Price Prediction of Airline Ticket Pricing Factors"

A

project report

submitted in partial

fulfilment of the requirements

for the award of the degree of

M.Sc. in Applied Statistics

SUBMITTED BY

BOTADARA VISHAL BHARATBHAI

(ROLL NO:18)

UNDER THE GUIDANCE OF

Ms. MANSI BALVA

DEPARTMENT OF STATISTICS

SARDAR PATEL UNIVERSITY

VALLABH VIDYANAGER, ANAND – 388120

GUJARAT (INDIA)

April-2026

CERTIFICATE

This is to certify that **Mr. VISHAL BHARATBHAI BOTADARA,** Roll No. 18, student of Master of Science in Applied Statistics, has satisfactorily completed his project work on **"A Comprehensive Statistical Analysis and Price Prediction of Airline Ticket Pricing Factors"** under the Paper Project Work **(**PS04CAST55**)** for M.Sc. Applied Statistics Semester IV during the period Nov 2025 – April 2026.

Place: Department of Statistics, V.V Nagar, Anand

Date: 4th April 2026

Ms. Mansi Balva	**Dr. Dharmesh Raykundaliya**
(Project Guide)	**(Head of Department)**

DECLARATION

I hereby declare that the report entitled "**A Comprehensive Statistical Analysis and Price Prediction of Airline Ticket Pricing Factors**" is an original work carried out by me -Botadara Vishal Bharatbhai under the guidance and supervision of Ms. Mansi Balva. This work has not been submitted, either in full or in part, to this or any other University for the award of any degree, diploma, or any other academic qualification.

I further declare that all the sources of information used in this study have been properly acknowledged.

Place: Department of Statistics, V.V Nagar, Anand
Date: 4th April 2026

Signature
Botadara Vishal Bharatbhai

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Ms. Mansi M. Balva, for his valuable guidance, continuous support, and insightful suggestions throughout the course of this project. His mentorship has been instrumental in the successful completion of this work.

I am highly thankful to Prof. Dr. Dharmesh P. Raykundaliya, Head of the Department, Department of Statistics, Sardar Patel University, for providing the necessary academic environment and encouragement to carry out this study.

I would also like to extend my sincere thanks to Prof. Dr. Jyoti M. Divecha, Dr. Khimaya Tinani, and Mr. Mahesh Bhingikar for their guidance in helping me understand and effectively apply statistical and machine learning techniques in real-life contexts. I am equally grateful to Mr. Viraj M. Tur and Ms. Kalyani Patil for their constant support and encouragement throughout this work.

I am also thankful to my classmates for their valuable discussions, support, and collaborative learning, which contributed to the successful completion of this project. I would also like to acknowledge the non-teaching staff of the department for their assistance and cooperation.

Finally, I express my grateful gratitude to my family for their unwavering support, encouragement, and understanding, which have been a constant source of strength throughout this journey.

Botadara Vishal Bharatbhai

Place: Dept. of Statistics, V.V Nagar, Anand

Date: 4th April 202

TABLE OF CONTENT

ABSTRACT

This project aims to analysis and predict outcomes using a structured dataset through the application of statistical and machine learning techniques. The primary objective is to identify key factors influencing the target variable and to develop accurate predictive models. The study begins with comprehensive Exploratory Data Analysis (EDA) to understand data distributions, detect patterns, and identify relationships among variables.

Data preprocessing plays a crucial role in the project, including handling missing values, removing inconsistencies, encoding categorical variables, and applying feature scaling techniques. These steps ensure that the dataset is clean, reliable, and suitable for model development.

To achieve the predictive objective, multiple supervised learning models are implemented, including Linear Regression, Decision Tree, Random Forest, and Logistic Regression (where applicable). Each model is trained and evaluated using appropriate performance metrics such as R-squared, Mean Squared Error (MSE), Accuracy, Precision, and Recall, depending on the nature of the problem.

The results of the study highlight the most influential features affecting the target variable and demonstrate the comparative performance of different models. Model optimization techniques are also applied to improve accuracy and reliability.

In conclusion, this project provides a comprehensive framework for data analysis and predictive modelling, emphasizing the importance of preprocessing, model evaluation, and interpretation in solving real-world problems using data-driven approaches.

INTRODUCTION

Background of study

In today's world, data plays an important role in decision-making across different fields. Organizations collect large amounts of data but often struggle to extract useful insights from it. Proper data analysis helps in understanding patterns and relationships within the data. However, raw data is usually unstructured and requires preprocessing before analysis. This makes data analysis an essential step in solving real-world problems.

Predictive modelling is widely used to estimate future outcomes based on historical data. Techniques such as regression and classification help in identifying important factors affecting the target variable. Before building models, data cleaning, encoding, and scaling are performed to improve accuracy. Exploratory Data Analysis (EDA) is also used to understand data distribution and trends. These steps ensure that the models are reliable and effective.

This study focuses on applying data analysis and machine learning techniques to a real dataset. The objective is to identify key factors and build accurate predictive models. Different models are evaluated using appropriate performance metrics. The results help in understanding the problem more clearly and improving decision-making. Overall, the study demonstrates the practical use of data-driven approaches.

Problem Statement

The dataset used in this project contains multiple features related to the target variable (such as price/outcome), but the relationships between these variables are not clearly defined. Factors like date, time, categorical attributes, and numerical inputs may influence the target, making the prediction task complex. Additionally, the dataset includes issues such as missing values, inconsistent formats, and categorical data that require proper preprocessing before analysis.

The key problem is to analysis this dataset to identify the most important factors affecting the target variable and to build accurate predictive models. Without proper analysis and modelling, it is difficult to understand how different features contribute to the outcome or to make reliable predictions.

Therefore, this project aims to clean and preprocess the dataset, perform detailed exploratory analysis, and apply suitable supervised machine learning models to improve prediction accuracy and generate meaningful insights for better decision-making.

Objectives of study

1.To perform exploratory data analysis (EDA) to understand the distribution, trends, and patterns in airline ticket prices.

2.To analysis the impact of key factors such as airline, class and Routes on ticket prices.

3.To examine statistical relationships between variables using techniques like correlation and hypothesis testing.

4.To apply quantile regression to study how different factors influence low, median, and high ticket prices.

5.To develop a Gamma regression model for accurate prediction of airline ticket prices considering skewed data.

6.To build and compare machine learning models to improve prediction accuracy.

Scope of the study

This study focuses on analyzing airline ticket pricing using a structured dataset containing variables such as airline, source and destination cities, travel class, duration, number of stops, departure time, and days left before departure. The scope of the study includes understanding how these factors influence ticket prices through comprehensive Exploratory Data Analysis (EDA) and visualization techniques.

The study further involves the application of statistical methods to examine relationships and differences within the data. Since the dataset is not normally distributed, non-parametric statistical tests such as the Kruskal-Wallis test, Mann-Whitney U test, and Spearman correlation are used to analyze pricing variations across different categories and to study relationships between key variables like duration, stops, and days left with ticket prices. Additionally, the Kolmogorov-Smirnov test is applied to assess the normality of the data.

In addition to statistical analysis, the scope extends to predictive modelling using both machine learning and statistical approaches. Models such as Decision Tree, Random Forest, XGBoost, Gamma Regression, and Quantile Regression are developed and evaluated using performance metrics like R^2 score and RMSE. The study also includes feature importance analysis to identify the most influential factors affecting ticket prices and residual analysis to assess model performance.

However, the scope is limited to the available dataset and does not include external real-world factors such as seasonal demand, fuel prices, airline policies, or economic conditions, which may also influence ticket pricing. The findings are therefore specific to the dataset used and are primarily intended for analytical understanding and academic purposes.

Limitation of the Study

This study is limited to the dataset used, which may not fully represent all real-world airline pricing scenarios. Important external factors such as seasonal demand, weather conditions, fuel prices, competition, and sudden market changes are not included in the dataset, which may affect the accuracy of the analysis and predictions.

The quality of the results depends on the quality of the data. Issues such as missing values, data inconsistencies, and limited feature availability may impact the performance of the models. Additionally, the study relies only on selected variables like airline, route, duration, stops, and booking time, which may not capture all factors influencing ticket prices.

Furthermore, the machine learning models used in this study are based on historical data and may not always perform well in dynamic real-time environments. The results are specific to this dataset and cannot be generalized to all airline pricing systems. Therefore, predictions should be interpreted with caution and within the context of the given data.

LITERATURE REVIEWS

1. Introduction to Airline Pricing Studies

Airline ticket pricing has attracted significant attention from researchers due to its dynamic and complex behaviour. Unlike fixed pricing systems, airline ticket prices fluctuate continuously based on multiple factors such as demand, route, travel class, booking time, and competition among airlines. According to various studies, pricing strategies in the airline industry are largely influenced by revenue management systems, which aim to maximize profit by adjusting prices in real time.

Earlier research highlights that ticket prices tend to increase as the departure date approaches, especially when demand is high. Conversely, early booking often results in lower prices. This dynamic pricing behaviour makes it challenging to predict ticket prices accurately. Researchers have therefore focused on identifying patterns in historical data to better understand pricing mechanisms.

2. Role of Exploratory Data Analysis in Pricing

Exploratory Data Analysis (EDA) plays a crucial role in understanding airline pricing datasets. Many studies emphasize the importance of visualizing data distributions and identifying relationships between variables before applying predictive models. Techniques such as histograms, box plots, and correlation matrices are widely used to detect trends, outliers, and feature importance.

Research has shown that variables such as **duration of flight, number of stops, airline type, and travel class** significantly influence ticket prices. For instance, longer flights and premium classes generally have higher prices, while flights with multiple stops may have lower prices compared to direct flights. EDA helps in uncovering such insights, which are essential for building accurate predictive models.

3. Impact of Key Variables on Ticket Pricing

Several researchers have analysis the impact of individual variables on airline ticket prices. One of the most important factors is the **number of days left before departure**, which has a strong inverse or direct relationship with price depending on demand conditions. Studies indicate that ticket prices often rise as the departure date approaches due to increased demand and limited seat availability.

Another important factor is **departure and arrival time**. Flights during peak hours or convenient timings (such as morning or evening) tend to be more expensive. Similarly, **airline brand and service quality** also influence pricing, with premium airlines charging higher fares compared to low-cost carriers.

The **number of stops** is also a key determinant of price. Direct flights are usually more expensive due to convenience, while flights with one or more stops are relatively cheaper. Additionally, route popularity and distance also play a major role in determining ticket prices.

4. Statistical Approaches in Airline Pricing Analysis

Statistical methods have been widely used to analysis airline pricing patterns. Researchers often apply hypothesis testing techniques to compare ticket prices across different categories such as airlines, classes, and routes. Since airline pricing data often does not follow a normal distribution, non-parametric tests such as the Mann-Whitney U test and Kruskal-Wallis test are commonly used.

These tests help in determining whether there are statistically significant differences in ticket prices between groups. For example, studies have shown significant price differences between economy and business class tickets, as well as among different airlines. Statistical analysis provides a strong foundation for understanding pricing behaviour before applying machine learning techniques.

5. Machine Learning Techniques for Price Prediction

With the advancement of data science, machine learning has become a popular approach for predicting airline ticket prices. Various supervised learning algorithms have been applied in previous studies, including Linear Regression, Decision Trees, and Random Forest.

Linear Regression is often used as a baseline model due to its simplicity and interpretability. However, it assumes a linear relationship between variables, which may not always hold true in real-world data. Decision Trees are capable of capturing non-linear relationships and interactions between variables, making them more suitable for complex datasets.

Random Forest, an ensemble learning method, has been widely recognized for its superior performance in prediction tasks. It combines multiple decision trees to improve accuracy and reduce overfitting. Studies consistently show that Random Forest models outperform traditional models in airline price prediction due to their robustness and ability to handle large datasets.

6. Model Evaluation and Performance Metrics

Evaluating model performance is a critical aspect of predictive modelling. Researchers commonly use metrics such as **Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2)** to assess regression models. These metrics help in determining how well the model predicts ticket prices compared to actual values.

Studies have shown that proper feature selection and hyperparameter tuning significantly improve model performance. Cross-validation techniques are also used to ensure that models generalize well to unseen data. The choice of evaluation metrics depends on the nature of the problem, but combining multiple metrics provides a more comprehensive assessment.

7. Challenges in Airline Price Prediction

Despite advancements in machine learning, predicting airline ticket prices remains challenging due to the presence of multiple external factors. Variables such as weather conditions, fuel prices, market competition, and sudden changes in demand are often not included in datasets but have a significant impact on pricing.

Additionally, airline pricing strategies are dynamic and may change frequently, making it difficult for models trained on historical data to adapt to real-time conditions. Data quality issues such as missing values, inconsistencies, and limited features further complicate the analysis.

8. Summary of Literature

The existing literature clearly demonstrates that airline ticket pricing is influenced by a combination of temporal, operational, and market-related factors. Exploratory Data Analysis helps in identifying key patterns, while statistical methods provide insights into group differences. Machine learning models, particularly ensemble methods like Random Forest, have proven to be effective in predicting ticket prices.

This study builds upon previous research by integrating data preprocessing, exploratory analysis, statistical testing, and machine learning techniques. The goal is to provide a comprehensive understanding of pricing behaviour and to develop accurate predictive models that can assist in decision-making.

Data Overview

The dataset used in this project contains over 300,000 airline ticket records, with the main goal of analyzing and predicting ticket prices. The target variable is price (₹), which represents the cost of flights. The data includes important flight-related details such as airline, source and destination cities, travel duration, number of stops, and booking time.

It consists of both categorical variables (airline, class, departure time, arrival time, cities) and numerical variables (duration, days_left, stops). These features help in understanding how different factors influence ticket pricing. The combination of these variables allows for detailed analysis of travel patterns and pricing behaviour.

Overall, the dataset is large, diverse, and reflects real-world airline pricing scenarios, including skewness and outliers. This makes it suitable for applying statistical methods and machine learning models to generate accurate predictions and meaningful insights.

DATASET AND VARIABLE EXPLANATIONS

Data Source

The dataset used in this project is obtained from a publicly available online source, specifically from the Kaggle platform. Kaggle is a well-known platform that provides a wide range of datasets for data analysis, machine learning, and research purposes.

The dataset contains detailed information about airline ticket bookings, including variables such as airline name, source and destination cities, departure and arrival time, duration of the journey, number of stops, class of travel, days left before departure, and ticket price. This data is suitable for analyzing pricing patterns and building predictive models.

Data Link : **https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data**

Dataset Description

The dataset used in this study consists of airline ticket booking information and is obtained from the Kaggle platform. It contains structured data related to various factors that influence airline ticket prices and is suitable for both statistical analysis and machine learning modelling.

The dataset includes multiple features such as airline name, source city, destination city, departure time, arrival time, duration of the journey, number of stops, travel class, days left before departure, and ticket price. These variables are categorized into independent variables (input features) and a dependent variable (target), where the ticket price is the main outcome to be predicted. The dataset contains both categorical variables (e.g., airline, class, source, destination) and numerical variables (e.g., duration, days left, price), requiring appropriate preprocessing techniques.

The dataset is well-suited for analysis pricing patterns and understanding how different factors affect ticket prices. It allows for performing Exploratory Data Analysis (EDA), statistical testing, and building predictive models. However, the dataset does not include external influencing factors such as seasonal demand, fuel prices, or real-time market conditions, which may also impact airline pricing.

DATASET

1	index	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	class	Duration	days_left	price
2	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Economy	130	1	5953
3	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	140	1	5953
4	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai	Economy	130	1	5956
5	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai	Economy	135	1	5955
6	4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai	Economy	140	1	5955
7	5	Vistara	UK-945	Delhi	Morning	zero	Afternoon	Mumbai	Economy	140	1	5955
8	6	Vistara	UK-927	Delhi	Morning	zero	Morning	Mumbai	Economy	125	1	6060
9	7	Vistara	UK-951	Delhi	Afternoon	zero	Evening	Mumbai	Economy	130	1	6060
10	8	GO_FIRST	G8-334	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	130	1	5954
11	9	GO_FIRST	G8-336	Delhi	Afternoon	zero	Evening	Mumbai	Economy	135	1	5954
12	10	GO_FIRST	G8-392	Delhi	Afternoon	zero	Evening	Mumbai	Economy	135	1	5954
13	11	GO_FIRST	G8-338	Delhi	Morning	zero	Afternoon	Mumbai	Economy	140	1	5954
14	12	Indigo	6E-5001	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	130	1	5955
15	13	Indigo	6E-6202	Delhi	Morning	zero	Afternoon	Mumbai	Economy	130	1	5955
16	14	Indigo	6E-549	Delhi	Afternoon	zero	Evening	Mumbai	Economy	135	1	5955
17	15	Indigo	6E-6278	Delhi	Morning	zero	Morning	Mumbai	Economy	140	1	5955
18	16	Air_India	AI-887	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	125	1	5955
19	17	Air_India	AI-665	Delhi	Early_Morning	zero	Morning	Mumbai	Economy	130	1	5955
20	18	AirAsia	I5-747	Delhi	Evening	one	Early_Morning	Mumbai	Economy	735	1	5949

Variable Explanations

1. Airline
Represents the airline operating the flight (e.g., IndiGo, Air India). Different airlines have different pricing strategies based on service quality and brand value.

2. Source City
Indicates the city from which the journey starts. Ticket prices may vary depending on demand and connectivity of the origin city.

3. Destination City
Represents the city where the journey ends. Prices differ based on route popularity, distance, and travel demand.

4. Departure Time
Shows the time of flight departure (morning, afternoon, evening, night). Flights during peak hours are usually more expensive.

5. Arrival Time
Indicates the time when the flight reaches the destination. Convenient arrival times may influence higher ticket pricing.

6. Duration
Represents the total travel time of the flight. It affects pricing as shorter and direct flights are often more expensive.

7. Stops
Indicates the number of stops between source and destination. Non-stop flights are generally costlier than flights with stops.

8. Class
Shows the travel class such as Economy or Business. Business class tickets are usually much more expensive.

9. Days Left
Represents the number of days remaining before departure. Ticket prices typically increase as the departure date approaches.

10. Price
The total cost of the flight ticket. It is the target variable that the study aims to analyze and predict.

METHODOLOGY

4.1Exploratory Data Analysis (EDA)

Introduction

Exploratory Data Analysis (EDA) is a fundamental step in the data analysis process used to **summarize, visualize, and understand the structure of the dataset** before applying statistical or machine learning models.

The main objective of EDA is to identify:

- Patterns and trends
- Missing values
- Outliers
- Relationships between variables

Dataset Understanding

Key Steps:

- Loaded dataset into a data frame
- Checked structure using:
    - head() → preview data
    - info() → data types and non-null counts
    - shape → number of rows and columns

Purpose:

To gain an initial understanding of the dataset and identify potential issues.

Data Cleaning

Handling Missing Values

- Missing values were identified using:
    - .isnull().sum()

Techniques Used:

- Numerical features → Mean/Median imputation
- Categorical features → Mode imputation or removal

Removing Duplicates

- Duplicate rows were identified and removed using:
    - .duplicated()

Data Type Conversion

- ❖ Converted columns (e.g., date, duration) into appropriate formats
- ❖ Ensured consistency across dataset

Univariate Analysis

Univariate analysis focuses on analyzing individual variables.

Numerical Variables

Techniques:

- ❖ Descriptive statistics:
  - Mean, Median, Standard Deviation
- ❖ Visualization:
  - Histogram
  - Boxplot

Insights:

- ❖ Distribution shape (normal/skewed)
- ❖ Presence of outliers
- ❖ Central tendency

Categorical Variables

Techniques:

- ❖ Frequency distribution (value_counts())
- ❖ Bar plots

Insights:

- ❖ Most frequent categories
- ❖ Class imbalance

Bivariate Analysis

Bivariate analysis examines relationships between two variables.

Numerical vs Numerical

Techniques:

- ❖ Scatter plots
- ❖ Correlation analysis

Insights:

- Positive/negative relationships
- Linear or non-linear patterns

Categorical vs Numerical

Techniques:

- Boxplots
- Grouped statistics

Insights:

- Impact of categorical variables on target
- Differences between groups

Multivariate Analysis

Multivariate analysis studies relationships among multiple variables.

Techniques:

- Correlation matrix
- Pair plots

Purpose:

- Identify relationships across multiple features
- Detect multicollinearity

Correlation Analysis

Correlation analysis was performed using:

- **Spearman Rank Correlation**

Reason:

- Data was non-normal
- Presence of outliers

Outcome:

- Identified strong and weak relationships
- Selected important features

Correlation Heatmap

A **Correlation Heatmap** was used to visualize relationships.

Purpose:

- Identify highly correlated variables

- Detect multicollinearity

Interpretation:

- Dark colour → strong correlation
- Light colour → weak correlation

Outlier Detection

Methods Used:

- Boxplots
- Interquartile Range (IQR)

Action Taken:

- Outliers were either:
    - Removed
    - Capped

Benefit:

Improved model performance and stability

Feature Engineering

New features were created to enhance predictive power.

Examples:

- Extracted:
    - Day, Month from date
    - Hour from time
- Converted duration into numeric format

Feature Transformation

Techniques:

- Log transformation for skewed data
- Scaling (if required)

Purpose:

- Normalize distribution
- Improve model performance

Encoding of Categorical Variables

Techniques:

- Label Encoding
- One-Hot Encoding

Reason:

Machine learning models require numerical input

Data Distribution Analysis

- Checked skewness and kurtosis
- Identified right-skewed variables

Outcome:

- Supported use of:
    - Gamma Model
    - Quantile Regression

4.2 Normality testing

- **Kolmogorov–Smirnov (KS) Test**

Definition

The **Kolmogorov–Smirnov Test** is a non-parametric statistical test used to compare:

- A sample distribution with a theoretical distribution (**one-sample KS test**), or
- Two sample distributions (**two-sample KS test**)

It measures the **maximum difference between cumulative distribution functions (CDFs)**.

Objective

To determine whether:

- A dataset follows a specific distribution (e.g., normal distribution), or
- Two datasets come from the same distribution

Types of KS Test

1. One-Sample KS Test

- Compares sample data with a known distribution

2. Two-Sample KS Test

- Compares two independent samples

Formula

The KS test statistic is defined as:

$$D = \sup_x | F_n(x) - F(x) |$$

Where:

- D= KS statistic (maximum distance)
- $F_n(x)$= empirical cumulative distribution function (ECDF)
- $F(x)$= theoretical cumulative distribution function
- sup= supremum (maximum difference)

For Two-Sample Test

$$D = \sup_x | F_1(x) - F_2(x) |$$

Where:

- $F_1(x), F_2(x)$= ECDFs of two samples

Interpretation of Statistic

- Larger D→ greater difference between distributions
- Smaller D→ distributions are similar

Hypothesis

One-Sample KS Test:

- **H_0:** Data follows the specified distribution
- **H_1:** Data does not follow the distribution

Two-Sample KS Test:

- **H_0:** Both samples come from the same distribution
- **H_1:** Distributions are different

Assumptions

- ❖ Data is **continuous**
- ❖ Observations are **independent**
- ❖ Distribution is fully specified (for one-sample test)
- ❖ No parameter estimation bias (important for accuracy)

Working Principle

- Compute ECDF of sample data
- Compare with theoretical CDF (or another ECDF)
- Calculate maximum vertical difference
- Evaluate significance using p-value

Interpretation

- If **p-value < 0.05** → Reject H_0 → Distributions differ
- If **p-value ≥ 0.05** → Fail to reject H_0

Applications

- Testing normality of data
- Comparing two datasets
- Model validation
- Distribution fitting
- Financial risk analysis
- Quality control

Advantages

- Non-parametric (no distribution assumption)
- Simple to compute
- Works for small sample sizes

Limitations

- Less sensitive at distribution tails
- Requires continuous data
- Not ideal when parameters are estimated from data

4.3 Mann–Whitney U Test

Definition

The **Mann–Whitney U Test** is a non-parametric statistical test used to compare **two independent groups** when the data does not follow a normal distribution.

Objective

To determine whether there is a **significant difference between two independent groups** based on their distributions.

Hypothesis

- **Null Hypothesis (H_0):**
  Both groups have the same distribution (no difference)
- **Alternative Hypothesis (H_1):**
  The distributions of the two groups are different

Formula

The test statistic is calculated as:

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Where:

- n_1= number of observations in group 1
- n_2= number of observations in group 2
- R_1= sum of ranks of group 1

Similarly, another statistic:

$$U' = n_1 n_2 - U$$

The **smaller value of $U$and U'**is used for testing.

Assumptions of Mann–Whitney U Test

- ❖ Independent observations
- ❖ Independent groups
- ❖ Ordinal or continuous dependent variable
- ❖ Similar distribution shape across groups
- ❖ Random sampling

Working Principle

- ❖ Combine both groups
- ❖ Assign ranks to all observations

- Calculate rank sums
- Compute U statistic
- Compare with critical value or p-value

When to Use

- Two independent groups
- Non-normal data
- Ordinal or continuous data

Interpretation

- If **p-value < 0.05** → Reject H_0 → Significant difference
- If **p-value ≥ 0.05** → Fail to reject H_0

Example (Project Use)

- Comparing **prices between two airlines**
- Comparing **treatment vs control group**

Advantages

- No normality assumption
- Robust to outliers
- Works with small samples

Limitations

- Only compares two groups
- Does not indicate magnitude of difference

4.4 Kruskal–Wallis Test

Definition

The **Kruskal–Wallis Test** is a non-parametric method used to compare **three or more independent groups**.

Objective

To determine whether at least **one group differs significantly** from others.

Hypothesis

- **Null Hypothesis (H_0):**
  All groups have the same distribution
- **Alternative Hypothesis (H_1):**
  At least one group differs

Formula

The Kruskal–Wallis statistic is:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

Where:

- N= total number of observations
- k= number of groups
- R_i= sum of ranks of group i
- n_i= number of observations in group i

Assumptions of Kruskal–Wallis Test

- ❖ Independent observations
- ❖ Independent groups
- ❖ Ordinal or continuous dependent variable
- ❖ Similar distribution shape across groups
- ❖ No requirement of normality

Working Principle

1. Combine all groups
2. Rank all observations

3. Compute rank sums for each group
4. Calculate H statistic
5. Compare with chi-square distribution

When to Use

- Three or more groups
- Non-normal data
- Independent samples

Interpretation

- If **p-value < 0.05** → Reject H_0 → At least one group differs
- If **p-value ≥ 0.05** → No significant difference

If significant → Perform **post-hoc tests** (e.g., Dunn's test)

Example (Project Use)

- Comparing **ticket prices across multiple airlines**
- Comparing **multiple treatment groups**

Advantages

- Works with multiple groups
- No normality assumption
- Suitable for skewed data

Limitations

- Does not identify which groups differ
- Requires post-hoc analysis

4.5 Correlation Analysis

1. Spearman Rank Correlation

Definition

The **Spearman Rank Correlation** is a **non-parametric measure** used to assess the **strength and direction of a monotonic relationship** between two variables.

Unlike Pearson correlation, it does **not assume linearity or normal distribution**.

Objective

To measure how well the relationship between two variables can be described using a **monotonic function** (increasing or decreasing).

Formula

The Spearman correlation coefficient (ρ) is calculated as:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

Where:

- d_i= difference between ranks of each observation
- n= number of observations

Steps to Calculate

1. Rank the values of both variables
2. Calculate the difference in ranks (d_i)
3. Square the differences (d_i^2)
4. Apply the formula

Range of Values

Value of ρ	**Interpretation**
+1	Perfect positive correlation
0	No correlation
-1	Perfect negative correlation

When to Use

- Data is **non-normal**

- Relationship is **monotonic (not necessarily linear)**
- Presence of **outliers**

Interpretation

- **ρ > 0** → Positive relationship
- **ρ < 0** → Negative relationship
- **ρ ≈ 0** → No relationship

Example (Project Context)

- Relationship between **duration and price**
- Relationship between **stops and ticket cost**

Advantages

- Works with ordinal data
- Robust to outliers
- No assumption of normality

Limitations

- Only detects monotonic relationships
- Less sensitive to linear relationships than Pearson

2. Heatmap (Correlation Heatmap)

Definition

A **Correlation Heatmap** is a graphical representation of correlation values using colors, helping to quickly identify relationships between multiple variables.

Objective

To visually display the **correlation matrix** and identify:

- Strong relationships
- Weak relationships
- Multicollinearity

How It Works

- Each cell represents correlation between two variables
- Colours indicate strength:
    - Dark colour → strong correlation
    - Light colour → weak correlation

Types of Correlation Used

- Pearson (linear)
- Spearman (rank-based) *(used in this project)*

Interpretation

- **High positive correlation (close to +1)**
  → Variables increase together
- **High negative correlation (close to -1)**
  → One increases, other decreases
- **Near zero**
  → No relationship

Example Insights

- Strong correlation between **duration and price**
- Moderate correlation between **number of stops and cost**
- Weak correlation between unrelated features

Importance in Project

- Helps in **feature selection**
- Detects **multicollinearity**
- Improves model performance

Advantages

- Easy visualization of complex data

- Quick identification of relationships
- Useful for high-dimensional datasets

Limitations

- Does not imply causation
- Can be misleading if misinterpreted

4.6 Quantile Regression

Definition

The **Quantile Regression** is a statistical technique used to estimate the **conditional quantiles (such as median, quartiles)** of a dependent variable.

Unlike linear regression, which estimates the **mean**, quantile regression estimates **different points of the distribution**, providing a more comprehensive analysis.

Objective

To understand how independent variables influence **different levels (quantiles)** of the response variable rather than only the average.

Model Representation

Quantile Regression Equation

$$Q_Y(\tau \mid X) = X\beta(\tau)$$

Where:

- $Q_Y(\tau \mid X)$: Conditional quantile of Y
- τ: Quantile ($0 < \tau < 1$)
- X: Independent variables
- $\beta(\tau)$: Coefficients for each quantile

Loss Function

Quantile regression minimizes the **check (pinball) loss function**:

$$\min_{\beta} \sum \rho_\tau (y_i - x_i\beta)$$

Where:

$$\rho_\tau(u) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0 \end{cases}$$

Interpretation of Loss

- Penalizes **underestimation and overestimation differently**
- For $\tau = 0.5$ → becomes **median regression**

Probability Density Function (PDF)

Quantile regression does not assume a fixed distribution like normal distribution. However, it is commonly associated with the:

- **Asymmetric Laplace Distribution**

Role of PDF:

- Provides a likelihood framework
- Helps estimate quantiles efficiently

Interpretation of Quantiles

Quantile	**Meaning**
$\tau = 0.25$	Lower values
$\tau = 0.50$	Median
$\tau = 0.75$	Higher values

Applications

- Airline Price Prediction
- Economics
- Healthcare
- Finance
- Environmental Studies

Advantages

- No normality assumption
- Robust to outliers

- Captures **heterogeneous effects**
- Works well with skewed data

Limitations

- Complex interpretation
- Computationally intensive
- Requires selection of quantiles

Comparison with Linear Regression

Feature	Linear Regression	Quantile Regression
Output	Mean	Quantiles
Assumption	Normality	No assumption
Outliers	Sensitive	Robust
Insight	Limited	Detailed

After Fitting the Model

After fitting the quantile regression model, several important steps must be performed:

1. Interpretation of Coefficients

- Analyze sign and magnitude of coefficients
- Check statistical significance (p-values)

Insight:

- Positive coefficient → increases quantile value
- Negative coefficient → decreases quantile value

2. Compare Multiple Quantiles

Fit models for:

- $\tau = 0.25$
- $\tau = 0.50$

- $\tau = 0.75$

Purpose:

- Understand how variable impact changes across distribution

3. Residual Analysis

Residual is:

$$\text{Residual} = y_i - \hat{y}_i$$

Check:

- Randomness of residuals
- Presence of patterns

Purpose:

- Validate model assumptions
- Detect bias

4. Goodness-of-Fit

Use:

- **Pseudo R^2**
- Quantile loss

Interpretation:

- Higher values indicate better fit

5. Visualization

Create:

- Predicted vs actual plots
- Quantile regression lines
- Residual plots

Purpose:

- Better understanding of model performance

6. Feature Importance

- Compare coefficients across quantiles

Insight:

- Some variables influence:
  - Low values more
  - High values more

7. Model Validation

Use:

- Train-test split
- Cross-validation

Metrics:

- Mean Absolute Error (MAE)
- Quantile loss

8. Interpretation

4.7 Gamma Model (Gamma Regression)

Definition

The **Gamma Regression** is a type of **Generalized Linear Model (GLM)** used when the response variable is **continuous, strictly positive, and right-skewed**.

It is especially useful when variance increases with the mean (heteroscedastic data).

Model Family

Gamma Regression belongs to the:

- **Generalized Linear Model (GLM)**

Components of GLM:

- **Random Component** → Gamma distribution
- **Systematic Component** → Linear predictor ($X\beta$)
- **Link Function** → Typically **log link**

Probability Density Function (PDF)

The Gamma distribution is defined as:

$$f(y;\alpha,\theta)=\frac{y^{\alpha-1}e^{-y/\theta}}{\Gamma(\alpha)\theta^{\alpha}},y>0$$

Where:

- y: response variable
- α: shape parameter
- θ: scale parameter
- $\Gamma(\alpha)$: gamma function

Mean and Variance

$$\text{Mean} = \mu = \alpha\theta$$
$$\text{Variance} = \alpha\theta^2$$

Variance increases with mean → suitable for skewed data

Model Formula

In Gamma regression with log link:

$$g(\mu) = \log(\mu) = X\beta$$
$$\mu = e^{X\beta}$$

Where:

- μ: expected value of response
- X: independent variables
- β: coefficients

Link Function

Commonly used link function:

- Log link:

$$\log(\mu) = X\beta$$

Other possible links:

- Identity link

- Inverse link

Assumptions

- ❖ Response variable is **positive (> 0)**
- ❖ Data is **right-skewed**
- ❖ Variance increases with mean
- ❖ Observations are independent

Applications (Only Names)

- ❖ Insurance claim modelling
- ❖ Healthcare cost analysis
- ❖ Airline ticket price prediction
- ❖ Survival analysis
- ❖ Risk modelling
- ❖ Financial loss modelling

Why Use Gamma Model?

- Handles **skewed continuous data**
- Suitable for **non-negative values**
- Models **heteroscedasticity** effectively

Interpretation of Coefficients

- Coefficients are interpreted in **multiplicative terms** due to log link

Example:

- If $\beta = 0.1$, then:
    - $e^{0.1} \approx 1.105$
    - Response increases by **~10.5%**

Estimation Method

- Parameters are estimated using **Maximum Likelihood Estimation (MLE)**

Advantages

- ❖ Suitable for skewed distributions
- ❖ Handles variance proportional to mean
- ❖ Flexible modelling framework

Limitations

- ❖ Cannot handle zero or negative values
- ❖ Sensitive to model misspecification
- ❖ Requires correct link function

Model Evaluation

Common evaluation metrics:

- ❖ Deviance
- ❖ AIC (Akaike Information Criterion)
- ❖ Log-likelihood

4.8 Machine Learning Models

Machine Learning

Definition

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables systems to automatically learn patterns from data and make predictions or decisions without being explicitly programmed. It focuses on building models that improve their performance as they are exposed to more data.

Types of Machine Learning

1. Supervised Learning

Supervised learning involves training a model using labeled data, where both input (features) and output (target) are known. The model learns the relationship between inputs and outputs and is used for prediction.

- **Types**:
    - Regression (continuous output) → e.g., price prediction
    - Classification (categorical output) → e.g., spam detection

- **Examples of Algorithms:**
    - Linear Regression
    - Decision Tree
    - Random Forest
    - XGBoost

2. Unsupervised Learning

Unsupervised learning deals with unlabelled data, where the model identifies hidden patterns or structures.

- **Types:**
    - Clustering → grouping similar data points
    - Association → finding relationships between variables
- **Examples**:
    - K-Means Clustering
    - Hierarchical Clustering
    - DBSCAN

3. Semi-Supervised Learning

This approach uses a combination of labelled and unlabelled data, useful when labelled data is limited.

4. Reinforcement Learning

In this type, the model learns by interacting with an environment and receiving rewards or penalties.

Applications of Machine Learning

Machine Learning is widely used in various domains:

- Airline Industry: Ticket price prediction, demand forecasting
- Healthcare: Disease prediction and diagnosis
- Finance: Fraud detection, credit scoring
- E-commerce: Recommendation systems

- Transportation: Route optimization and dynamic pricing

How Machine Learning Algorithms Work

Machine learning algorithms generally follow these steps:

1. Data Collection

Gather relevant data from sources such as databases or datasets.

2. Data Preprocessing

- Handling missing values
- Encoding categorical variables
- Feature scaling

3. Feature Selection

Selecting important variables that influence the target variable.

4. Model Training

The algorithm learns patterns from training data by minimizing error.

5. Prediction

The trained model predicts outcomes on new/unseen data.

6. Model Evaluation

Performance is measured using metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R^2 Score

Working Principle (General Idea)

Machine learning models try to learn a function:

$$Y = f(X) + \epsilon$$

Where:

- Y= Target variable (Price)
- X= Input features (Airline, Duration, etc.)
- ϵ= Error term

The goal is to minimize the error between actual and predicted values.

Conclusion

Machine Learning provides a powerful framework for modelling complex relationships in data. In this project, ML techniques are applied to capture the influence of multiple factors on airline ticket prices and to build predictive models that can accurately estimate ticket prices.

1. Linear Regression

What is Linear Regression?

Linear Regression is a **supervised machine learning algorithm used for regression problems**, where the target variable is numerical.

In your project:

Input: airline, class, duration, days_left, stops, source city, destination city, etc.
Output: predicted airline ticket price.

The basic idea is to find the best mathematical relationship between the input variables and ticket price.

Basic equation

Where:

- predicted ticket price
- intercept
- coefficients
- input features

Advantages

- Simple to understand.
- Fast.
- Easy to interpret.
- Good baseline model.
- Useful for understanding relationships between variables.

Limitations

- Assumes a linear relationship.
- Cannot easily capture complicated nonlinear relationships.
- Can be affected by outliers.
- May perform poorly when relationships between features and price are complex.

2. Ridge Regression

What is Ridge Regression?

Ridge Regression is an improved version of Linear Regression that uses **regularization**.

It is especially useful when there are many features or when features are correlated with each other.

Why do we need Ridge?

Suppose several variables contain similar information.

For example:

- stops_numeric
- is_direct
- One-hot encoded stops-related information

Some features may be correlated.

Ordinary Linear Regression can produce unstable or excessively large coefficients.

Ridge controls this problem by adding a **penalty**.

Mathematical idea

Linear Regression minimizes:

Ridge minimizes:

The second part is the **L2 regularization penalty**.

This controls the strength of regularization.

Higher alpha → stronger regularization.

Lower alpha → model becomes closer to ordinary Linear Regression.

Advantages

- Reduces overfitting.
- Handles multicollinearity better.
- Makes coefficients more stable.
- Useful when there are many features.

Limitations

- Still assumes a basically linear relationship.
- Doesn't naturally capture complex nonlinear patterns.
- Choosing the regularization strength is important.

3.Decision Tree

Definition

A Decision Tree is a supervised machine learning algorithm that makes predictions by repeatedly dividing the dataset into smaller groups based on decision rules.

It has a tree-like structure consisting of:

- Root node – the starting point of the tree.
- Decision/internal nodes – points where the data is split.
- Branches – represent the outcomes of a decision.
- Leaf nodes – final nodes containing the prediction.

How Decision Tree works

The algorithm works through the following process:

Step 1: Start with the complete training dataset

All observations are initially placed at the root node.

Step 2: Search for the best split

The algorithm evaluates possible splitting rules and selects the split that produces the greatest improvement in prediction quality.

For regression, measures such as Mean Squared Error (MSE) are commonly used to determine the quality of a split.

Step 3: Divide the dataset

The selected rule divides the observations into separate groups.

Step 4: Repeat the process

The same process is applied recursively to the resulting groups.

Step 5: Stop splitting

Splitting stops when a stopping condition is reached, such as maximum tree depth, minimum number of observations, or insufficient improvement.

Step 6: Make the prediction

For regression, the final prediction at a leaf is generally based on the average target value of observations reaching that leaf.

Main concept

The fundamental idea is:

Recursively divide the data into increasingly homogeneous groups until useful predictions can be made.

Advantages

- Simple and intuitive.
- Can model nonlinear relationships.
- Can capture complex interactions.
- Requires relatively little data preparation.
- Easy to interpret compared with many other algorithms.

Disadvantages

- Can easily overfit.
- Deep trees can become very complex.
- Small changes in training data can produce a very different tree.
- A single tree can have high variance.
- Generalization may be weaker than ensemble methods.

4.Random Forest

Definition

Random Forest is an ensemble supervised machine learning algorithm that combines the predictions of multiple Decision Trees.

The central idea is:

Instead of relying on one Decision Tree, build many different trees and combine their predictions.

For regression, the predictions of the individual trees are generally **averaged** to produce the final prediction.

How Random Forest works

Step 1: Create different training samples

Random Forest creates multiple datasets from the original training data using **bootstrap sampling**.

Each sample can contain a different combination of observations.

Step 2: Build Decision Trees

A separate Decision Tree is trained using each bootstrap sample.

Step 3: Introduce randomness in feature selection

During tree construction, only a randomly selected subset of available features is considered at each splitting point.

This makes the individual trees different from one another.

Step 4: Grow multiple trees

The process is repeated until the specified number of trees has been constructed.

Step 5: Generate predictions

Each tree independently produces a prediction.

Step 6: Combine the predictions

For regression, the predictions are averaged:

where is the number of trees.

Why does Random Forest work?

A single Decision Tree can be unstable and may overfit the training data.

Random Forest reduces this problem by combining many different trees.

If individual trees make different errors, averaging their predictions can reduce the overall variance.

Two important sources of randomness

Random Forest mainly introduces randomness through:

1. **Bootstrap samples of observations**
2. **Random subsets of features**

This creates diverse trees.

Main concept

Many diverse Decision Trees are combined to produce a more stable and reliable prediction.

Advantages

- Usually more stable than a single Decision Tree.
- Reduces variance.
- Can handle nonlinear relationships.
- Can capture complex interactions.
- Generally resistant to overfitting compared with a single tree.
- Works well with high-dimensional and tabular datasets.
- Can estimate feature importance.

Disadvantages

- More computationally expensive than one Decision Tree.

- Requires more memory.
- Less interpretable than a single tree.
- A large number of trees can increase training and prediction time.

5.Gradient Boosting

Definition

Gradient Boosting is an ensemble supervised machine learning algorithm that builds multiple weak prediction models **sequentially**.

The most important idea is:

Each new model tries to correct the errors made by the existing ensemble.

Decision Trees are commonly used as the weak learners in Gradient Boosting.

How Gradient Boosting works

Step 1: Create an initial prediction

The algorithm begins with a simple initial model.

For a regression problem, the initial prediction is commonly based on a suitable estimate of the target, often the mean.

Step 2: Calculate the errors

The algorithm compares the actual values with the predictions produced by the current model.

These errors indicate where the model needs improvement.

Step 3: Train a new weak learner

A new Decision Tree is trained to learn the errors or, more precisely, the **negative gradient of the loss function**.

This tree focuses on correcting the weaknesses of the current model.

Step 4: Add the new model

The new tree is added to the existing ensemble.

However, its contribution is controlled by the **learning rate**.

Step 5: Repeat

The algorithm repeatedly:

1. Makes predictions.
2. Measures the current errors.
3. Builds another weak learner.

4. Adds its correction.
5. Improves the overall model.

Step 6: Produce the final prediction

After all boosting iterations are completed, the predictions from all stages are combined to produce the final result.

RESULTS AND INTERPRETATION

5.1 Objective 1: To perform exploratory data analysis (EDA) to understand the distribution, trends, and patterns in airline ticket prices.

This objective focuses on performing Exploratory Data Analysis (EDA) to understand the overall structure and behaviour of airline ticket price data. It involves analyzing the distribution of prices to identify patterns such as skewness and outliers. Visualization techniques like histograms, boxplots, and scatter plots are used to detect trends and relationships among variables. EDA helps in identifying important factors that influence ticket pricing. It also provides a foundation for further statistical analysis and model building.

Distribution of Price(₹)

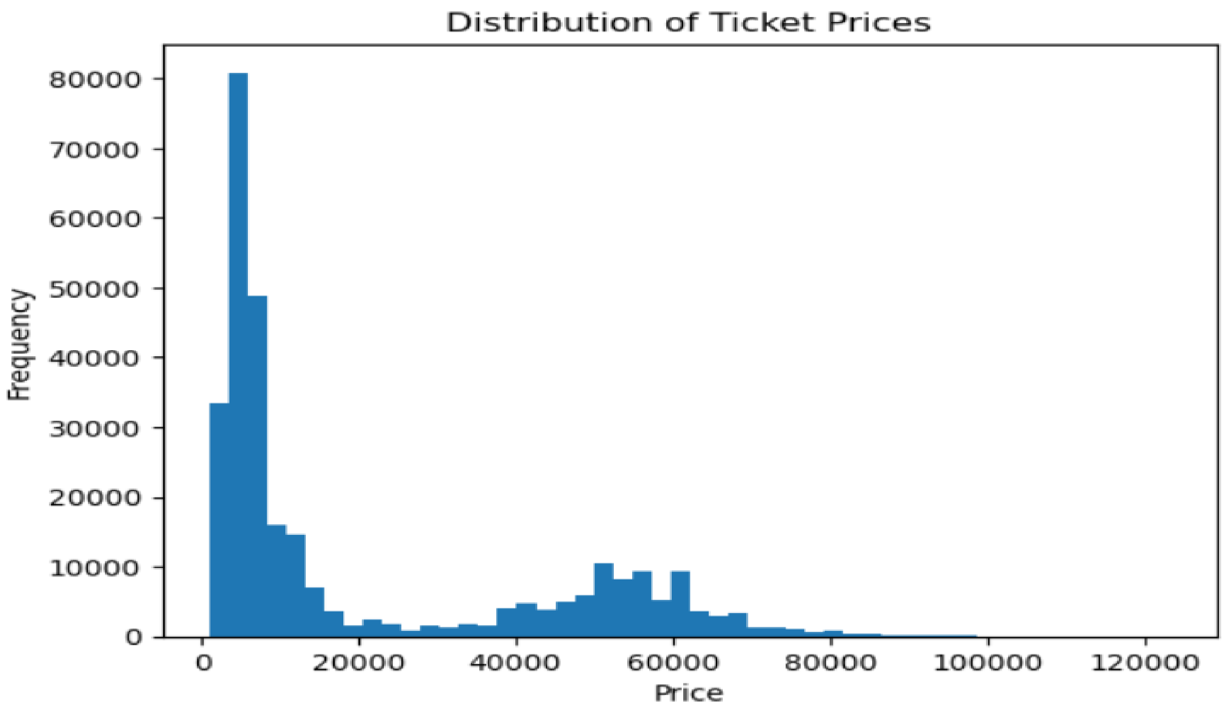


Fig.5.1.1 Distribution of Ticket Price

Interpretation : The histogram of ticket prices shows that the data is **positively skewed (right-skewed)**, meaning that most of the ticket prices are concentrated in the lower range, while a smaller number of tickets have very high prices. The majority of prices lie approximately between ₹2,000 and ₹15,000, indicating that most travellers opt for budget or economy flights. Additionally, there is a noticeable secondary concentration of prices in the higher range (around ₹40,000 to ₹70,000), which likely represents premium or business-class tickets, suggesting the presence of multiple pricing segments in the dataset. The distribution also includes a few extreme values reaching above ₹1,00,000, indicating outliers, possibly due to last-minute bookings or luxury travel. Overall, the data does not follow a normal distribution, and this justifies the use of advanced models such as quantile regression and Gamma regression, which are better suited for handling skewed and non-negative data.

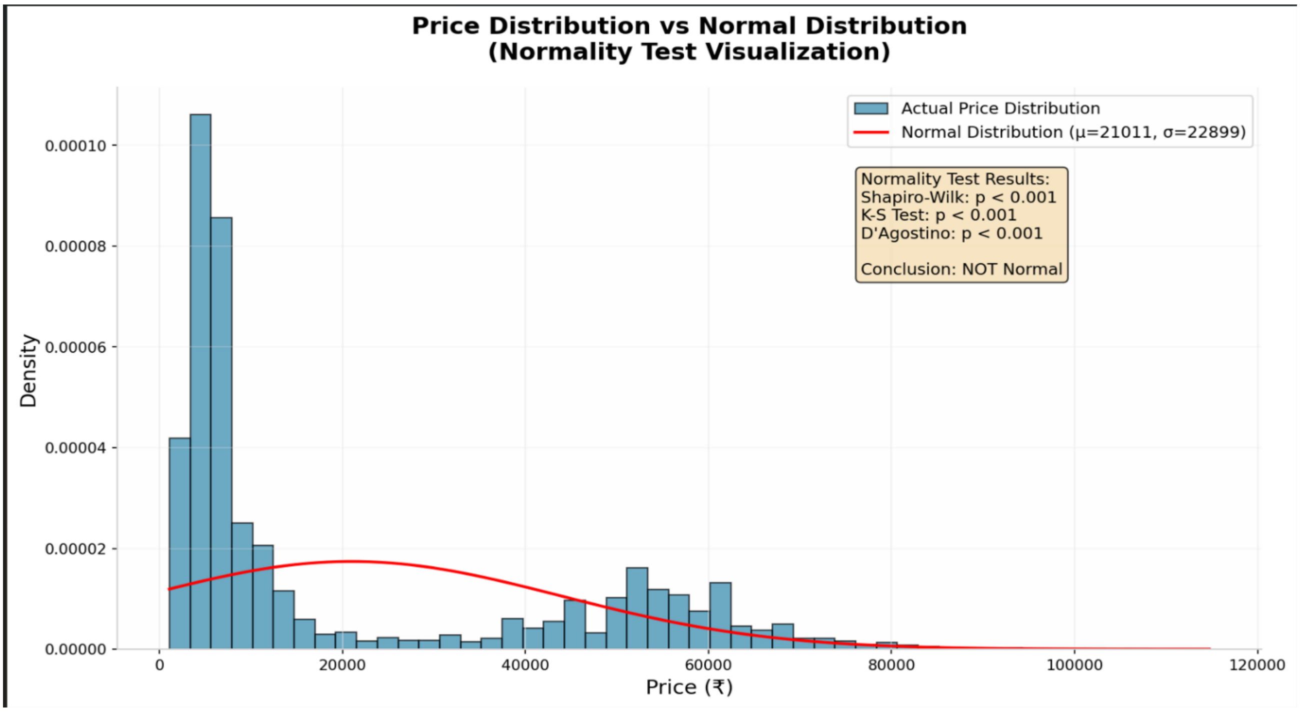


Fig.5.1.2 Price Distribution

Kolmogorov-Smirnov Test

Hypothesis:

- **Null Hypothesis (H_0):**
  The airline ticket price data follows a normal distribution.
- **Alternative Hypothesis (H_1):**
  The airline ticket price data does not follow a normal distribution.

KS Statistic: 0.2781726047890952

p-value: 0.00013

Interpretation:

The Kolmogorov–Smirnov (KS) test was conducted to assess whether the airline ticket price data follows a normal distribution. The test produced a KS statistic of 0.278 and a p-value of 0.00013. Since the p-value is significantly less than the standard significance level of 0.05, the null hypothesis is rejected. This indicates that there is a statistically significant difference between the observed distribution of ticket prices and a normal distribution. Therefore, it can be concluded that the airline ticket price data does not follow a normal distribution. This result supports the use of non-parametric and distribution-based models, such as quantile regression and Gamma regression, which are more appropriate for analyzing skewed data.

Stops And Day_Left Distribution Plot

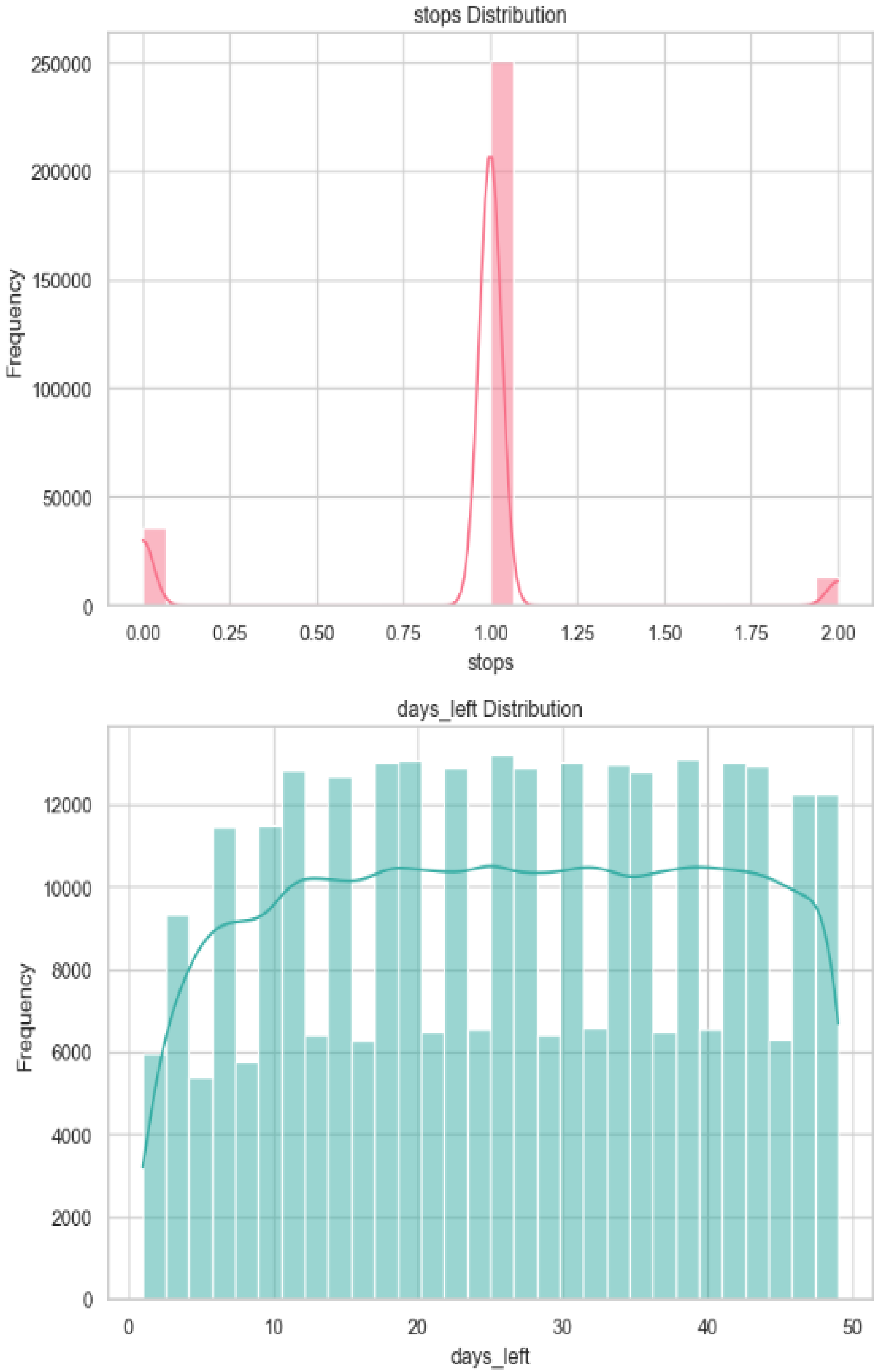


Fig.5.1.3 Stops And Day left Distribution

Interpretation:

The histogram of stops shows that the data is highly concentrated around 1 stop, indicating that most flights in the dataset involve a single layover. There are comparatively fewer non-stop (0 stops) and two-stop flights, making the distribution highly skewed and discrete rather than continuous. This suggests that one-stop flights dominate airline operations, which may influence both travel duration and ticket pricing patterns.

The histogram of days_left (days before departure) shows a relatively uniform distribution across the range of 1 to 50 days, with no strong peaks. This indicates that bookings are spread fairly evenly over time, rather than concentrated at a specific period. Such a pattern suggests that booking behaviour varies widely among customers, and days_left can be an important variable for capturing pricing dynamics, especially in relation to early and last-minute bookings.

Duration Distribution Plot

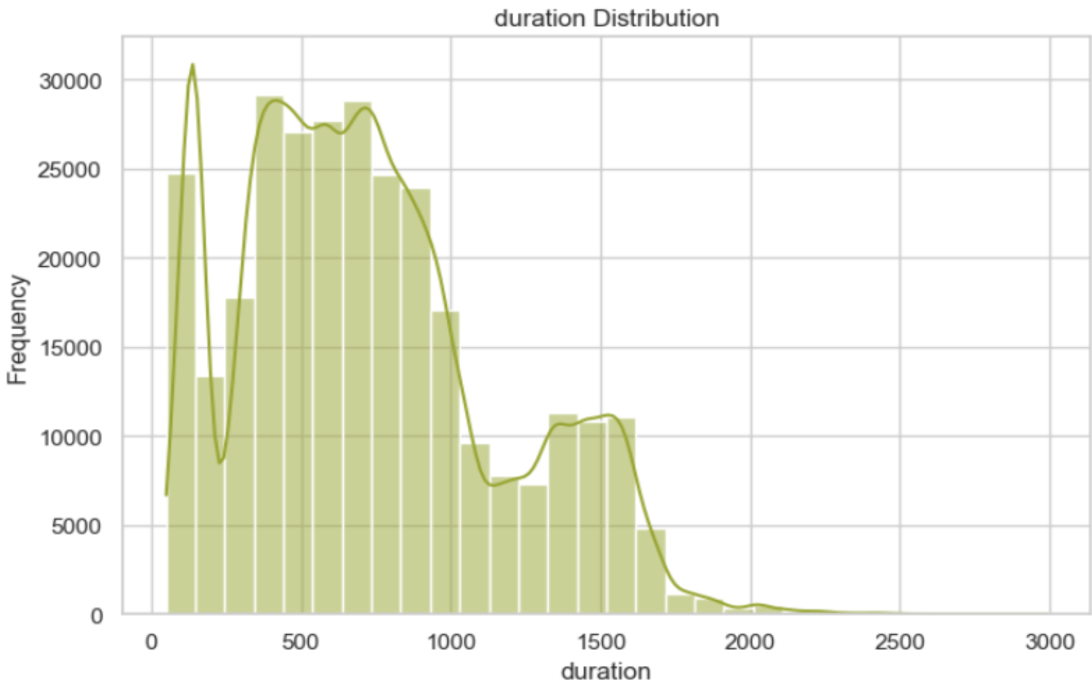


Fig.5.1.4 Duration Distribution

The distribution of duration is right-skewed, with most flights having shorter to moderate durations.
Most values are concentrated between 300 and 1000 minutes, indicating common medium-distance flights.
There is a long tail, showing the presence of long-duration or multi-stop flights (outliers).
The multimodal pattern suggests different categories of flights (short, medium, long durations).

Airlines Frequency Distribution

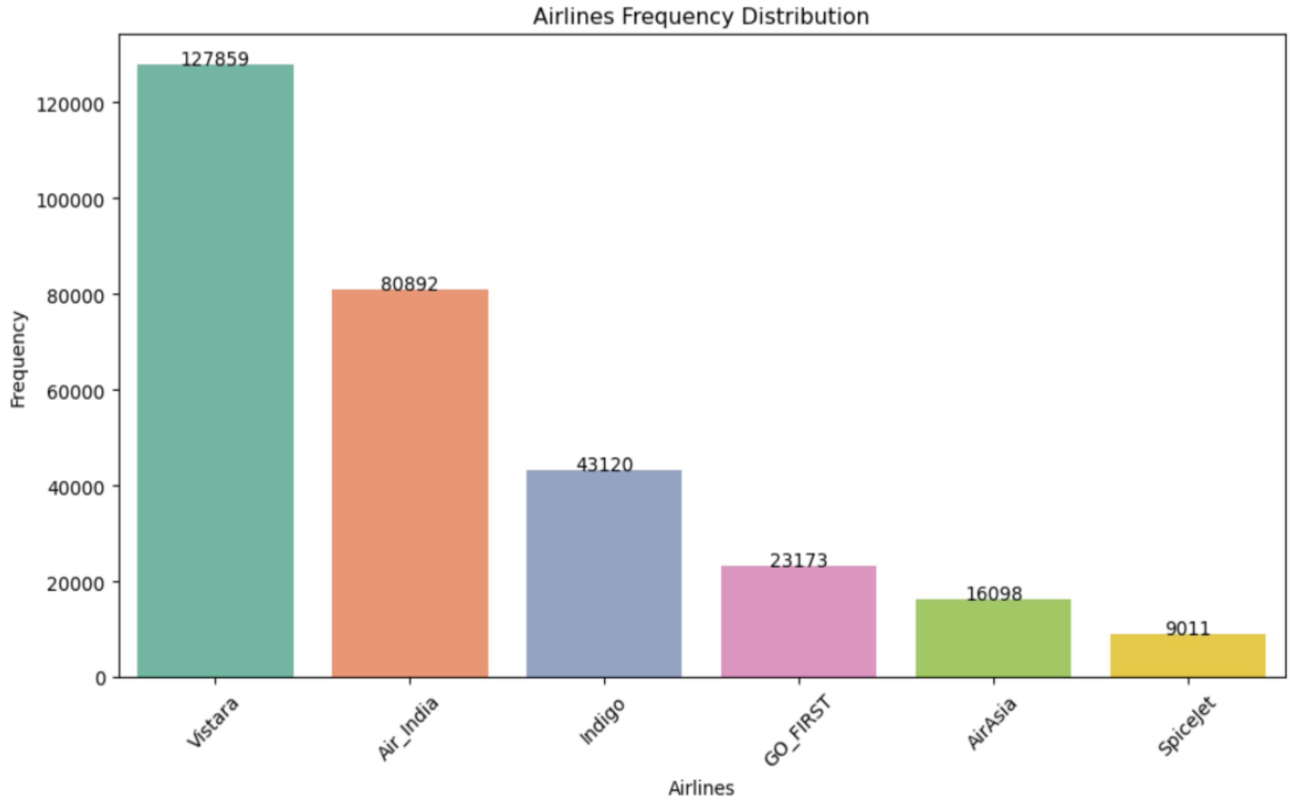


Fig.5.1.5 Airlines Frequency Bar chart

Interpretation:

The bar chart shows the distribution of flights across different airlines in the dataset. It is evident that Vistara has the highest number of flights (127,859), making it the most dominant airline in the dataset. This is followed by Air India (80,892) and Indigo (43,120), which also contribute a significant portion of the total flights.

On the other hand, airlines such as GO FIRST (23,173), AirAsia (16,098), and SpiceJet (9,011) have comparatively fewer flights. Among all airlines, SpiceJet has the lowest frequency, indicating it has the least representation in the dataset.

Overall, the distribution is imbalanced, with a few airlines dominating the dataset while others have relatively low representation. This imbalance may influence model predictions, as models might give more weight to airlines with higher frequency. Additionally, the variation in frequencies reflects real-world airline operations, where certain airlines operate more routes or have higher availability compared to others.

Count Plot of class

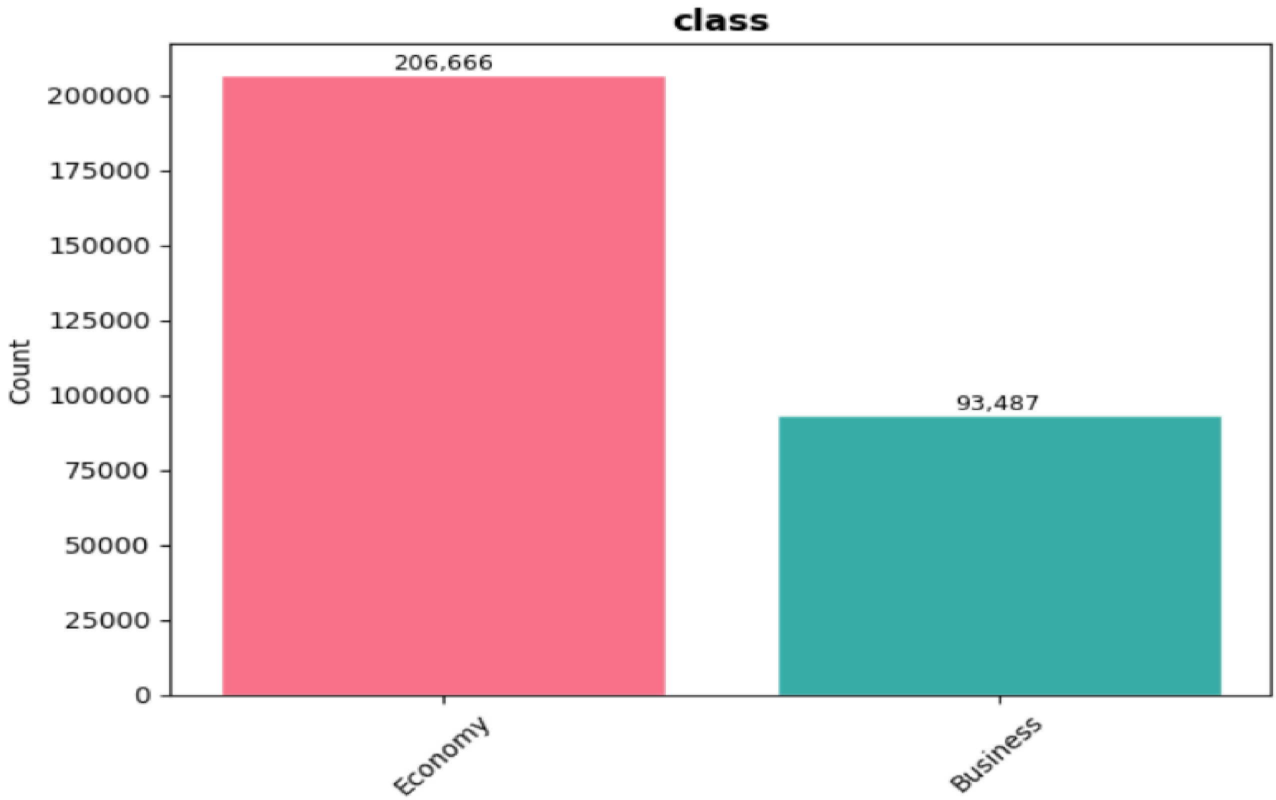


Fig.5.1.6 Class Bar Chart

Interpretation:

The bar chart shows the distribution of flight bookings across different travel classes. It is evident that the Economy class dominates the dataset with 206,666 observations, while the Business class has significantly fewer bookings with 93,487 observations. This indicates that a large majority of passengers prefer economy class over business class.

This imbalance suggests that economy travel is more common and accessible, likely due to its lower cost compared to business class. On the other hand, business class represents a smaller segment of travellers, typically associated with higher ticket prices and premium services.

From an analytical perspective, this uneven distribution may influence model outcomes, as the model may learn more patterns from economy class data due to its higher frequency. Additionally, this distribution reflects real-world travel behaviour, where most passengers prioritize cost-effective travel options over luxury.

Median Ticket Price by bar chart

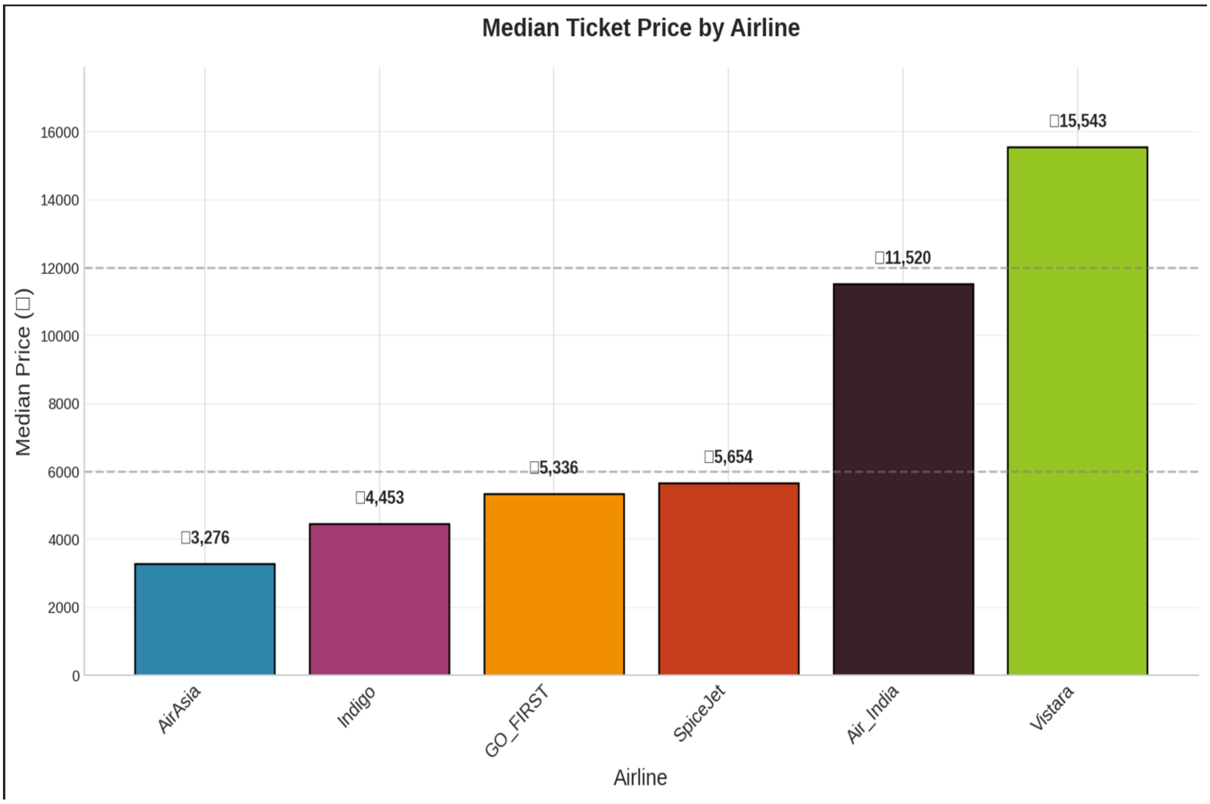


Fig.5.1.7 Median Ticket Price by bar chart

Interpretation:

The bar chart presents the median ticket prices across different airlines, highlighting clear differences in pricing strategies. It is observed that Vistara has the highest median ticket price (₹15,543), followed by Air India (₹11,520), indicating that these airlines are positioned in the premium segment. On the other hand, airlines such as SpiceJet (₹5,654), GO FIRST (₹5,336), and Indigo (₹4,453) fall in the mid-range category, offering relatively affordable options.

The lowest median price is observed for AirAsia (₹3,276), making it the most budget-friendly airline among all. This variation clearly reflects a segmentation in the airline market, where some airlines focus on low-cost travel while others provide premium services at higher prices.

Additionally, the noticeable gap between premium airlines (Vistara, Air India) and budget airlines suggests that airline choice has a significant impact on ticket pricing. This insight is important for both travellers and predictive modelling, as the airline variable plays a crucial role in determining flight prices.

Price Distribution by Airline

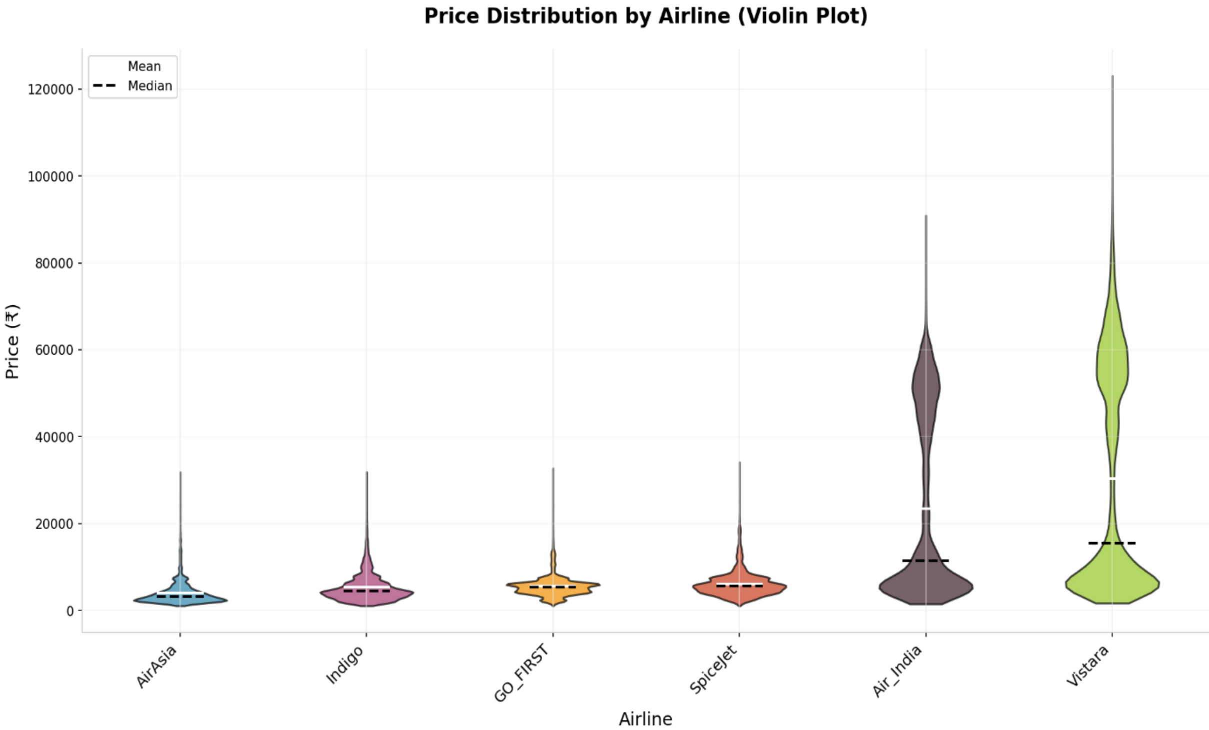


Fig.5.1.8 Price Distribution By Airline (violine plot)

Interpretation:

The violin plot shows the distribution of ticket prices for each airline, highlighting differences in pricing patterns. Airlines like AirAsia and Indigo have narrow and concentrated distributions at lower price ranges, indicating that their ticket prices are generally consistent and affordable. This suggests they operate mainly in the low-cost segment.

Airlines such as GO FIRST and SpiceJet display moderate spread in their price distributions, indicating some variability in ticket pricing. While they are still relatively affordable, their prices vary more compared to purely budget airlines, placing them in the mid-range segment.

In contrast, Air India and Vistara exhibit wide and highly spread distributions with long upper tails, indicating significant variability and the presence of high-priced tickets. This reflects their premium positioning in the market, offering both economy and higher-class services, resulting in a broader range of ticket prices.

Price Distribution Plot By Stops

Fig.5.1.9 Box plot of no. of stops

Interpretation:

The box plot shows how ticket prices vary based on the number of stops. It is observed that direct flights (0 stops) have the lowest median price (₹4,499), indicating that non-stop flights are generally cheaper in this dataset. Their price range is also relatively narrow, suggesting more consistent pricing.

Flights with one stop have a higher median price (₹7,959) and a very wide spread, indicating high variability in prices. The presence of extreme values (very high prices) suggests that some one-stop flights can be significantly more expensive, possibly due to airline type, route, or booking conditions.

For two or more stops, the median price (₹8,307) is slightly higher than one-stop flights, but the variability is lower compared to one-stop flights. Overall, the analysis suggests that ticket prices tend to increase with the number of stops, and flights with stops show greater variability and potential for higher prices compared to direct flights.

Boxplot of Price

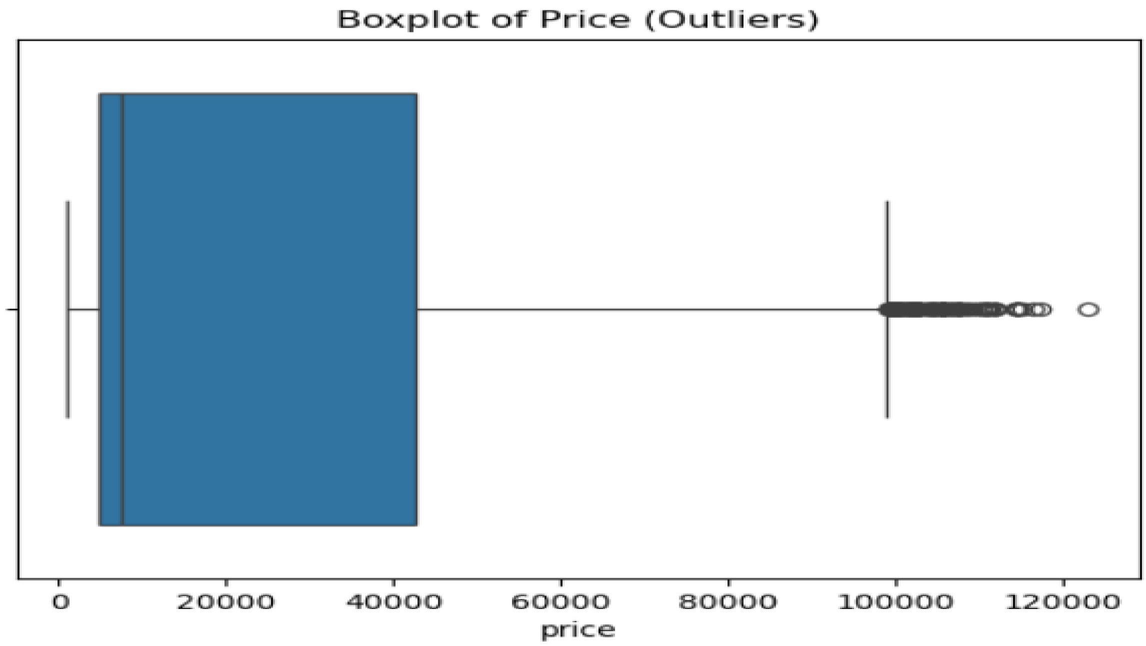


Fig.5.1.10 Boxplot of Price

The boxplot indicates that most ticket prices are concentrated in the lower range, with the median closer to the lower values. This shows that the majority of flights are relatively affordable. The distribution is right-skewed, as the spread of prices extends more toward higher values.

Additionally, several points beyond the upper whisker represent outliers, indicating the presence of extremely high ticket prices. These outliers may correspond to premium or last-minute bookings, confirming that the dataset contains significant variability and extreme values.

Skewness And Kurtosis

Skewness: 1.0613772532064343

Kurtosis: -0.3962927186960772

Interpretation:

The skewness value of 1.06 indicates that the data is positively skewed (right-skewed). This means most ticket prices are concentrated at lower values, while a smaller number of observations extend toward higher prices. In practical terms, cheaper tickets are more common, and expensive tickets occur less frequently but stretch the distribution to the right.

The kurtosis value of -0.396 suggests that the distribution is platykurtic, meaning it is slightly flatter than a normal distribution with lighter tails. This indicates fewer extreme outliers than a highly peaked distribution, although some high-value prices still exist. Overall, the data is moderately skewed with a relatively flat shape compared to a normal distribution.

Scatter Plot of Price V/s Duration

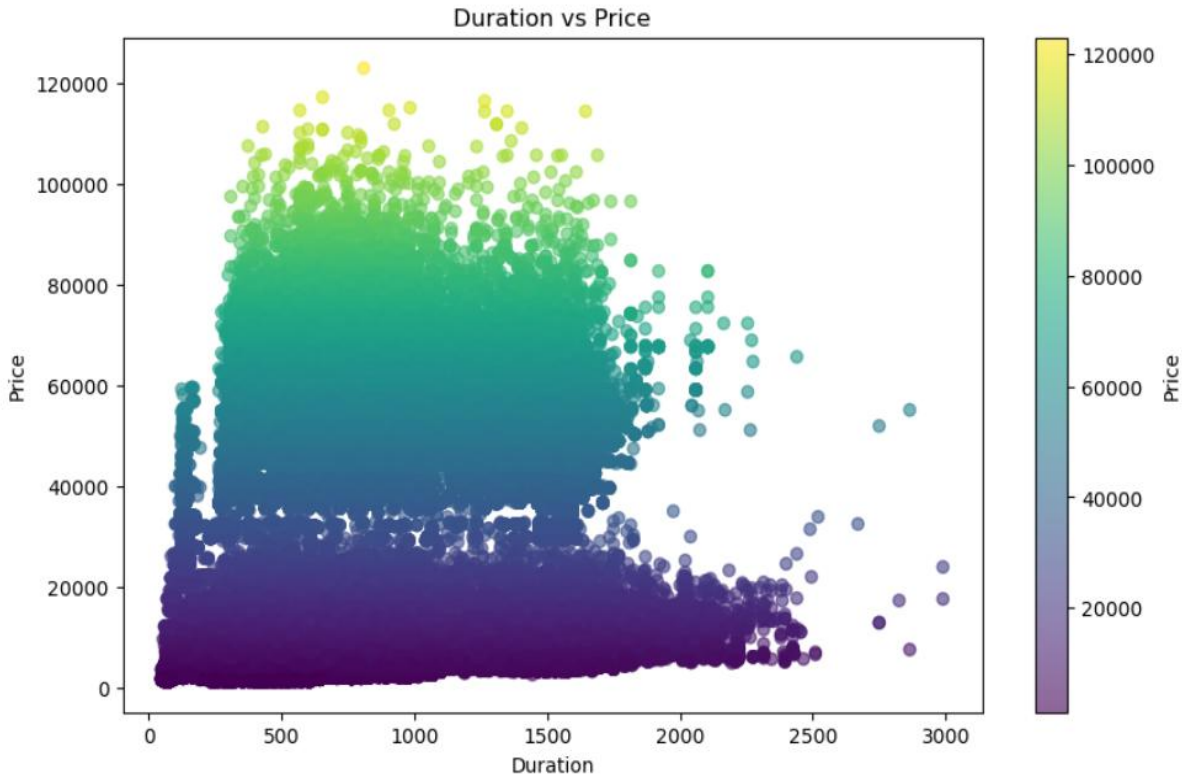


Fig.5.1.11 Duration v/s Price Scatter Plot

Interpretation:

The scatter plot shows the relationship between flight duration and ticket price, with colour intensity representing price levels. It is observed that there is no strong linear relationship between duration and price, as the data points are widely scattered. While some increase in price is visible with increasing duration, the pattern is not consistent, indicating that duration alone does not determine ticket price.

Additionally, the plot reveals distinct clusters of prices, where low-priced tickets are densely concentrated across all durations, and higher-priced tickets appear in specific regions. This suggests that other factors such as airline, class, and number of stops play a significant role in pricing. The presence of scattered high-price points also indicates variability and possible premium or outlier tickets.

Overall, the analysis shows that ticket price is influenced by multiple factors, and duration has only a partial impact, making it necessary to use multivariate models for accurate prediction

Scatterplot Day Left v/s Price

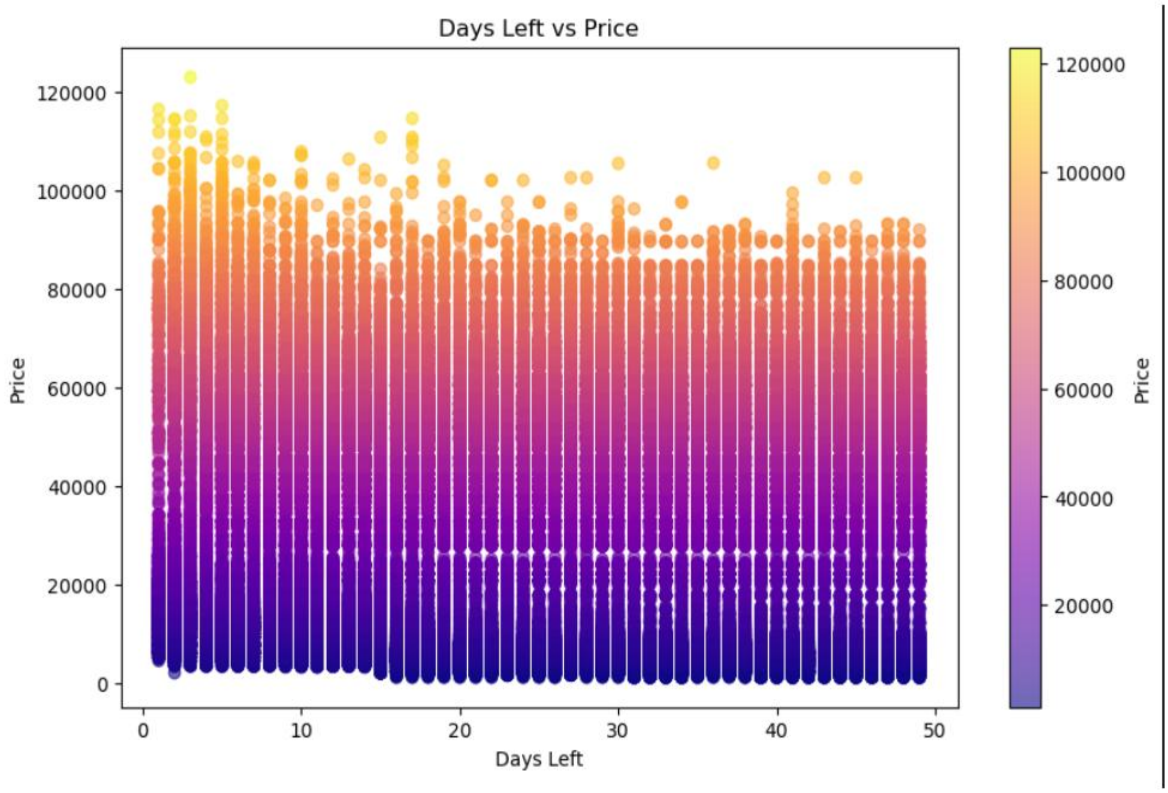


Fig.5.1.12 day Left v/s Price Scatter Plot

Interpretation:

The scatter plot shows the relationship between the number of days left before departure and ticket price, with colour indicating price levels. It is observed that prices tend to be higher when the number of days left is low, meaning tickets are more expensive when booked closer to the departure date. As the number of days left increases, lower-priced tickets become more common, indicating that early booking generally results in cheaper fares.

However, the plot also shows a wide spread of prices at all levels of days left, suggesting that days left alone does not fully determine ticket prices. High-priced tickets are still present even when booking early, indicating the influence of other factors such as airline, class, and number of stops. Overall, there is a negative relationship (inverse trend) between days left and price, but it is not perfectly linear.

Departure Time vs Price

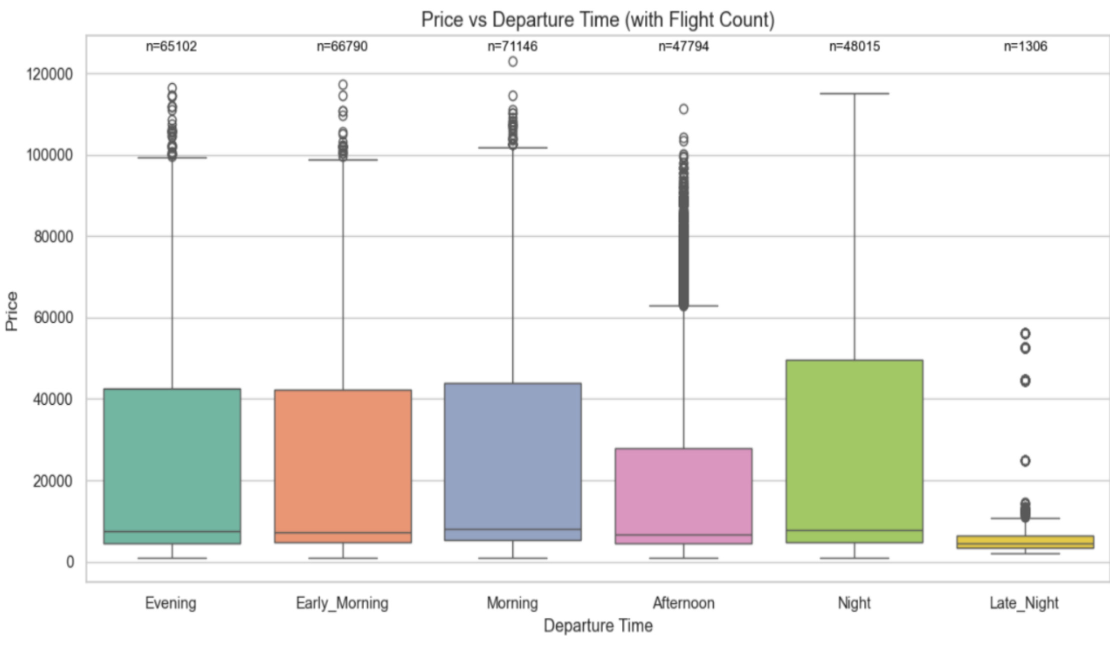


Fig.5.1.13 Departure Time Box Plot

Interpretation:

The boxplot shows the distribution of ticket prices across different departure times along with the number of flights in each category. It is observed that Morning, Early Morning, and Evening flights have the highest number of observations, indicating that these are the most commonly scheduled travel times. In contrast, Late Night flights have very few observations (n=1306), showing limited availability.

In terms of pricing, Night and Morning flights tend to have higher median prices and wider variability, indicating the presence of both low and very high ticket prices. Afternoon flights generally have comparatively lower median prices, suggesting they may be more economical options. Meanwhile, Late Night flights have the lowest and most consistent prices, with a narrow spread.

Overall, the analysis indicates that departure time influences ticket pricing, with peak travel times (Morning, Night) showing higher variability and potential for expensive tickets, while less preferred times like Late Night offer more stable and lower prices.

Correlation Heatmap

Fig.5.1.14 Heatmap of variables

Interpretation:

The heatmap shows the correlation between all variables in the dataset. It is observed that price has a strong negative correlation with class (-0.94), indicating that higher-class categories (like business) are associated with significantly higher ticket prices. Additionally, price has a moderate positive correlation with airline (0.24), flight (0.31), and duration (0.20), suggesting that these factors also influence ticket pricing to some extent.

Most other variables, such as departure_time, arrival_time, source_city, destination_city, and days_left, show very weak or negligible correlation with price, indicating that they have limited direct linear impact individually. However, this does not mean they are unimportant, as they may still influence price in combination with other variables.

There are also some strong correlations among independent variables, such as source_city and route (0.99) and stops with duration (0.47), which may indicate multicollinearity. Overall, the heatmap suggests that class is the most important factor affecting price, followed by airline, flight, and duration, while other variables have weaker individual relationships.

Source City Bar Chart

Fig.5.1.15 Source city Frequency Bar chart

Interpretation:

The bar chart shows the distribution of flights across different source cities. It is observed that Delhi (61,343) and Mumbai (60,896) have the highest number of flights, indicating that these cities are the major hubs for air travel in the dataset. Bangalore (52,061) also has a significant share, while Kolkata (46,347) falls in the mid-range.

On the lower side, Hyderabad (40,806) and Chennai (38,700) have comparatively fewer flights, suggesting lower traffic or fewer route options from these cities. Overall, the distribution indicates that flight availability is concentrated in major metropolitan cities, with Delhi and Mumbai dominating the network, which may also influence pricing and connectivity patterns.

Destination City Bar Chart

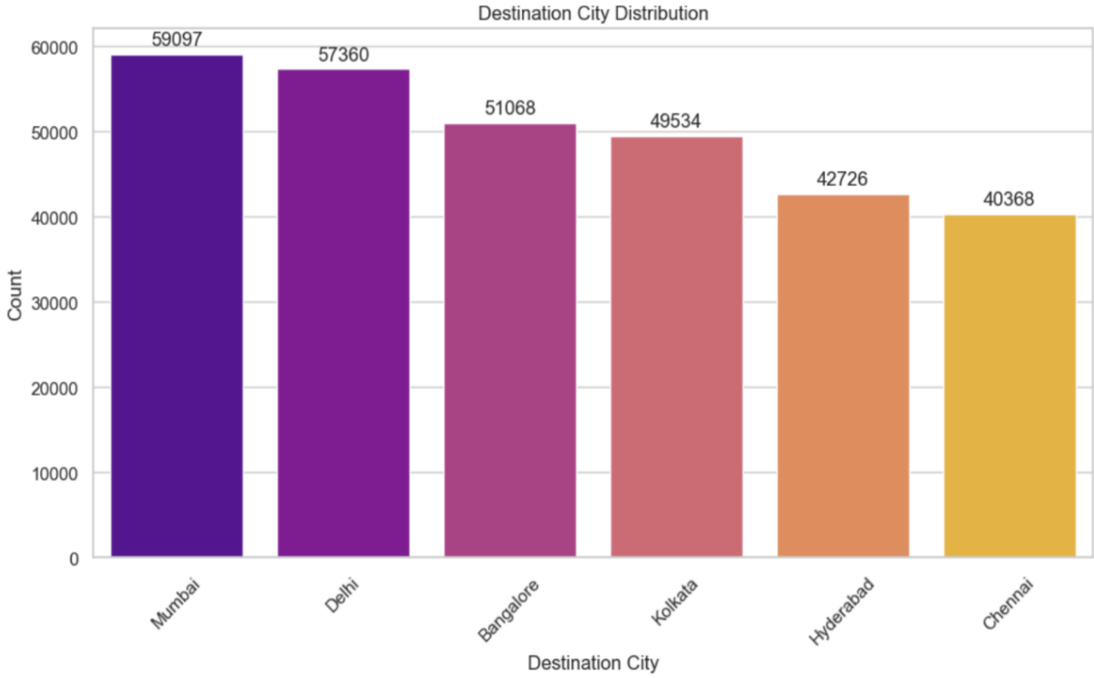


Fig.5.1.16 Destination City Bar Chart

Interpretation:

The bar chart shows the distribution of flights across destination cities. It is observed that Mumbai (59,097) and Delhi (57,360) receive the highest number of flights, indicating that these cities are the primary destinations and major air traffic hubs. Bangalore (51,068) and Kolkata (49,534) also have a substantial number of incoming flights, reflecting their importance in the network.

On the other hand, Hyderabad (42,726) and Chennai (40,368) have comparatively fewer incoming flights, suggesting relatively lower demand or connectivity. Overall, the distribution indicates that flight traffic is concentrated toward major metropolitan cities, especially Mumbai and Delhi, which likely play a key role in pricing, demand patterns, and route popularity.

Stops Bar Charts

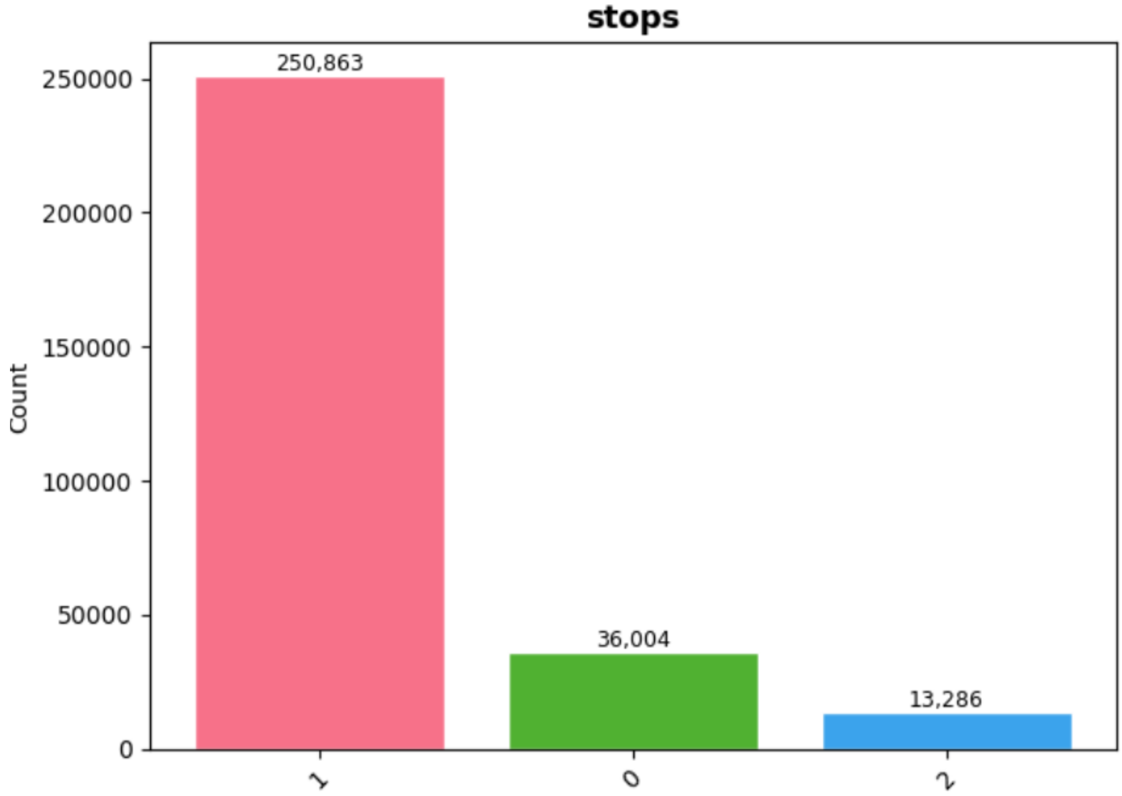


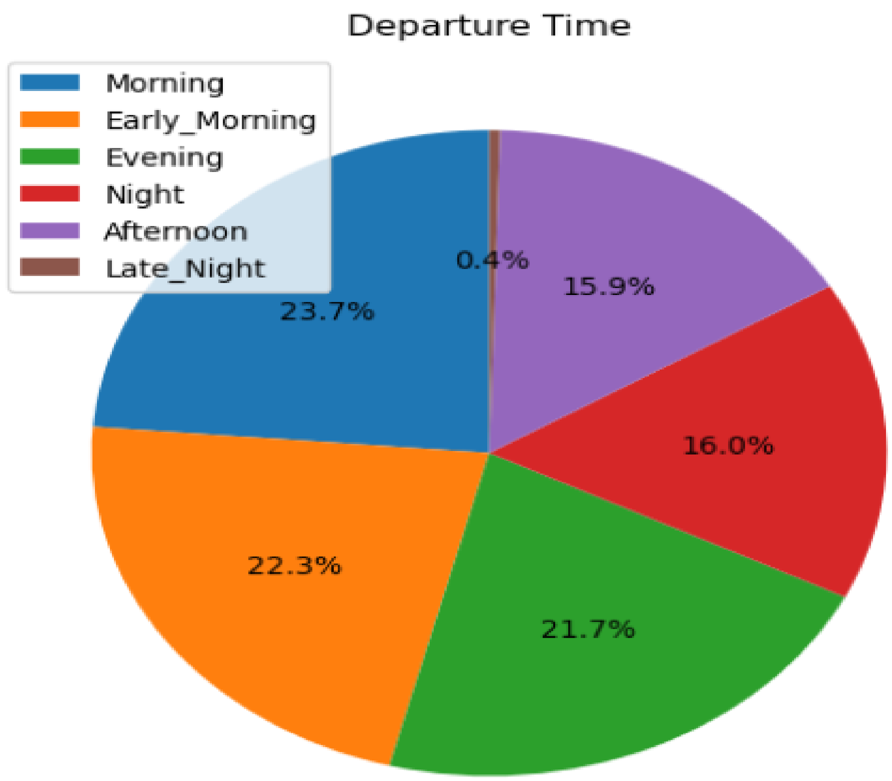
Fig.5.1.17 Stops Bar Chart

Interpretation:

The chart shows the distribution of flights based on the number of stops. It is observed that flights with 1 stop dominate the dataset (250,863), indicating that most airline routes are not direct and include at least one layover. In comparison, non-stop (0 stop) flights are significantly fewer (36,004), suggesting that direct flights are less common and possibly limited to major routes.

Flights with 2 stops (13,286) are the least frequent, indicating that longer, multi-stop journeys are relatively rare. Overall, the distribution highlights that one-stop flights are the most common travel option, which may impact both travel duration and ticket pricing, as flights with more stops are often associated with longer travel times and varying cost structures.

Pie Chart of Arrival And Departure Time



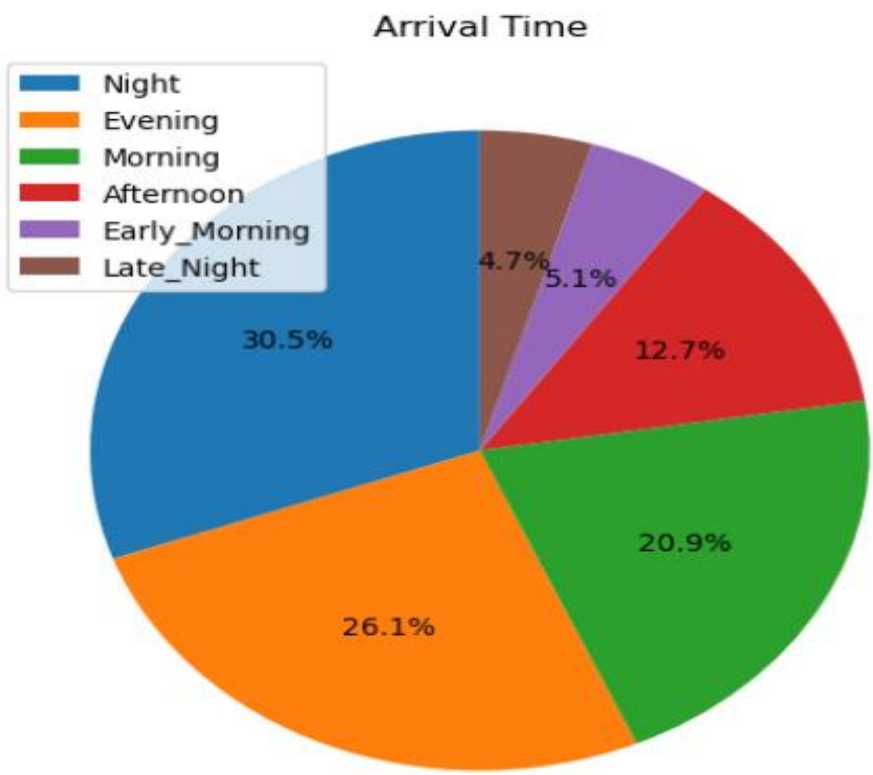


Fig 5.1.18Pie Chart of Departure Time Arrival Time

Interpretation : Departure Time Distribution

The pie chart shows that Morning (23.7%), Early Morning (22.3%), and Evening (21.7%) have the highest share of flight departures. This indicates that airlines schedule more flights during these peak hours to match high passenger demand, especially for business and routine travel.

Night (16.0%) and Afternoon (15.9%) have moderate proportions, while Late Night (0.4%) has a very small share. This suggests that late-night departures are least preferred due to inconvenience and lower demand.

Interpretation : Arrival Time Distribution

The arrival time chart shows that Night arrivals dominate (30.5%), followed by Evening (26.1%) and Morning (20.9%). This indicates that many flights are scheduled to arrive during night hours, likely due to longer travel durations and optimized scheduling.

In contrast, Afternoon (12.7%), Early Morning (5.1%), and Late Night (4.7%) have lower proportions. This suggests that fewer flights are scheduled to arrive during these less convenient or less demanded time periods.

DESCRIPTIVE STATISTICS

Overall Price Statistics

Statistic	Value
Count	300153
Mean	20889.66
Median	7425
Standard Deviation	22697.77
Variance	515188643
Minimum	1105
Maximum	123071
Range	121966
Q1 (25th percentile)	4783
Q3 (75th percentile)	42521
IQR	37738
Skewness	1.0614
Kurtosis	-0.3963

Price Statistics by Airline

Airline	Count	Mean (₹)	Median (₹)	Std Dev (₹)	Min (₹)	Max (₹)
AirAsia	16098	4091.07	3276	2824.06	1105	31917
Indigo	43120	5324.22	4453	3268.89	1105	31952
GO_FIRST	23173	5652.01	5336	2513.87	1105	32803
SpiceJet	9011	6179.28	5654	2999.63	1106	34158
Air India	80892	23507.02	11520	20905.12	1526	90970
Vistara	127859	30396.54	15543	25637.16	1714	123071

Price Statistics by Number of Stops

Stops	Count	Mean (₹)	Median (₹)	Std Dev (₹)	Min (₹)	Max (₹)
0	36004	9375.94	4499	10623.01	1105	59573
1	250863	22900.99	7959	23626.07	1105	123071
2	13286	14113.45	8307	17664.33	1966	117307

	count	mean	std	min	25%	50%	75%	max
stops	300153.0	0.924312	0.398106	0.0	1.0	1.0	1.0	2.0
duration	300153.0	733.259055	431.518381	50.0	410.0	675.0	970.0	2990.0
days_left	300153.0	26.004751	13.561004	1.0	15.0	26.0	38.0	49.0
price	300153.0	20889.660523	22697.767366	1105.0	4783.0	7425.0	42521.0	123071.0

Interpretation :

Price (Target Variable)

The ticket price shows high variability, with a mean of ₹20,889 and a median of ₹7,425, indicating a right-skewed distribution where a few very high prices increase the average. The large standard deviation (₹22,697) and wide range (₹1,105 to ₹123,071) confirm the presence of extreme values and outliers, making price a highly dispersed variable.

Duration

Flight duration has a moderate spread, with an average of around 733 minutes and a median of 675 minutes. The wide range (50 to 2990 minutes) suggests that flights vary significantly from short to very long journeys. This variability indicates that duration is an important factor influencing ticket price.

Days Left (Before Departure)

The average number of days left is about 26 days, with values ranging from 1 to 49 days. The relatively balanced distribution suggests that bookings are made across different time windows. This variable is important as it reflects booking behaviour and its impact on pricing, where earlier or later bookings may influence ticket cost.

Stops

The average value of stops is close to 1, indicating that most flights have one stop. The distribution shows that non-stop flights are fewer, while flights with more stops are less common. This variable plays a key role in pricing, as more stops generally increase travel time and may affect ticket prices.

Airline

There is a clear difference in pricing across airlines. Vistara and Air India have significantly higher average and median prices, indicating premium pricing, while AirAsia and Indigo offer lower fares, representing budget options. This shows that airline type is a strong determinant of ticket price.

Conclusion of Objective 1

The exploratory data analysis (EDA) provided a comprehensive understanding of the structure, distribution, and behaviour of airline ticket prices. The analysis revealed that ticket prices are highly skewed and non-normally distributed, with most prices concentrated in the lower range and a few extreme high values. This was statistically confirmed using the Kolmogorov–Smirnov test, supporting the need for advanced models such as Gamma and Quantile regression for accurate analysis .

Further analysis identified key factors influencing ticket prices, including airline, class, duration, stops, and days_left. Premium airlines like Vistara and Air India showed significantly higher prices, while budget airlines offered lower fares. Variables such as duration and number of stops contributed to price variation, whereas booking timing (days_left) showed an inverse relationship with price, indicating that early bookings tend to be cheaper.

Overall, the EDA highlighted important patterns, trends, and relationships within the dataset, forming a strong foundation for predictive modeling. It confirmed that ticket pricing is influenced by multiple interacting factors rather than a single variable, making it essential to use multivariate and robust statistical models for accurate prediction and decision-making.

5.2 Objective : 2 To analysis the impact of key factors such as airline ,Class and Routes on ticket prices.

Airline Wise

Airline	Median Price (₹)	Mean Price (₹)	Minimum Price (₹)	Maximum Price (₹)
Vistara	15,543	30,396.54	1,714	123,071
Air India	11,520	23,507.02	1,526	90,970
SpiceJet	5,654	6,179.28	1,106	34,158
GO_FIRST	5,336	5,652.01	1,105	32,803
Indigo	4,453	5,324.22	1,105	31,952
AirAsia	3,276	4,091.07	1,105	31,917

Class Wise

Class	Median Price (₹)	Mean Price (₹)	Minimum Price (₹)	Maximum Price (₹)
Business	53,164	52,540.08	12,000	123,071
Economy	5,772	6,572.34	1,105	42,349

Route Wise

Route	Median Price (₹)	Mean Price (₹)	Number of Flights
Chennai → Bangalore	10,469	25,081.85	6,493
Bangalore → Chennai	9,241	23,321.85	6,410
Kolkata → Chennai	8,589	23,660.36	6,653
Kolkata → Hyderabad	8,467	21,500.01	7,897
Chennai → Kolkata	8,394	22,669.93	6,983
Chennai → Mumbai	8,233	22,765.85	9,338
Mumbai → Chennai	8,148	22,781.90	10,130
Bangalore → Kolkata	8,112	23,500.06	10,028
Kolkata → Bangalore	8,111	22,744.81	9,824
Kolkata → Mumbai	7,958	22,078.88	11,467

Kruskal Wallis test (Airline Groups)

Comparison of Median Ticket Prices Across Different Airlines Using Kruskal–Wallis Test

Hypothesis

- Null Hypothesis (H_0):
  There is no significant difference in median ticket prices across different airlines.
- Alternative Hypothesis (H_1):
  There is a significant difference in median ticket prices across different airlines.

Test Statistic

- H-statistic: 94,612.81
- p-value: 0.000035

Interpretation

Since the p-value is less than the significance level (0.05), the null hypothesis is rejected. This indicates that there is a statistically significant difference in median ticket prices among different airlines. Therefore, airline type has a significant impact on ticket pricing, with some airlines offering higher median prices compared to others.

❖ Kruskal Wallis Test (Routes)

Comparison of Median Ticket Prices Across Different Routes Using Kruskal–Wallis Test

Hypothesis

- Null Hypothesis (H_0):
  There is no significant difference in median ticket prices across different routes.
- Alternative Hypothesis (H_1):
  There is a significant difference in median ticket prices across different routes.

Test Statistic

- H-statistic: 5,047.90
- p-value: 0.00

Interpretation

Since the p-value is less than the significance level (0.05), the null hypothesis is rejected. This indicates that there is a statistically significant difference in median ticket prices across different routes. Therefore, the route of travel significantly influences ticket pricing, suggesting that certain routes are consistently more expensive than others due to factors such as demand, distance, and connectivity.

❖ Mann–Whitney U Test (Compare Class)

Comparison of Ticket Prices Between Economy and Business Class Using Mann–Whitney U Test

Hypothesis

- Null Hypothesis (H_0):
  There is no significant difference in ticket prices between Economy and Business class.
- Alternative Hypothesis (H_1):
  There is a significant difference in ticket prices between Economy and Business class.

Test Statistic

- U-statistic: 6,234,631.50
- p-value: 0.000054

Interpretation

Since the p-value is less than the significance level (0.05), the null hypothesis is rejected. This indicates that there is a statistically significant difference in ticket prices between Economy and Business class.

The results show that Business class tickets have significantly higher prices compared to Economy class, confirming that class type is a major determinant of airline ticket pricing.

Class Pie Chart

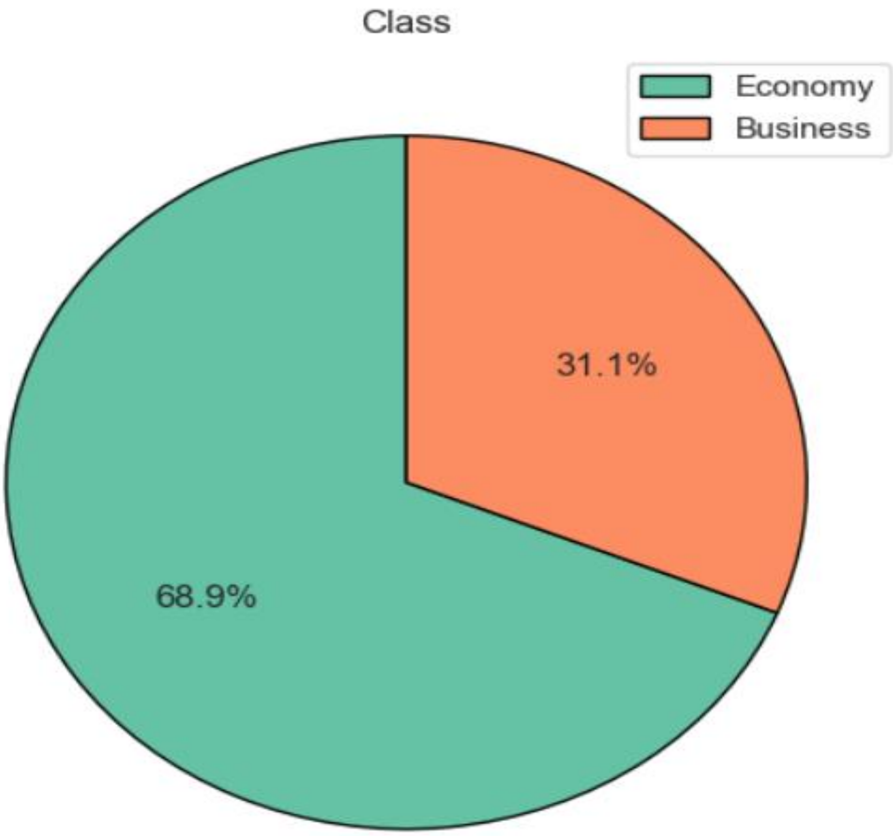


Fig.5.2.1 Pie Chart od Class

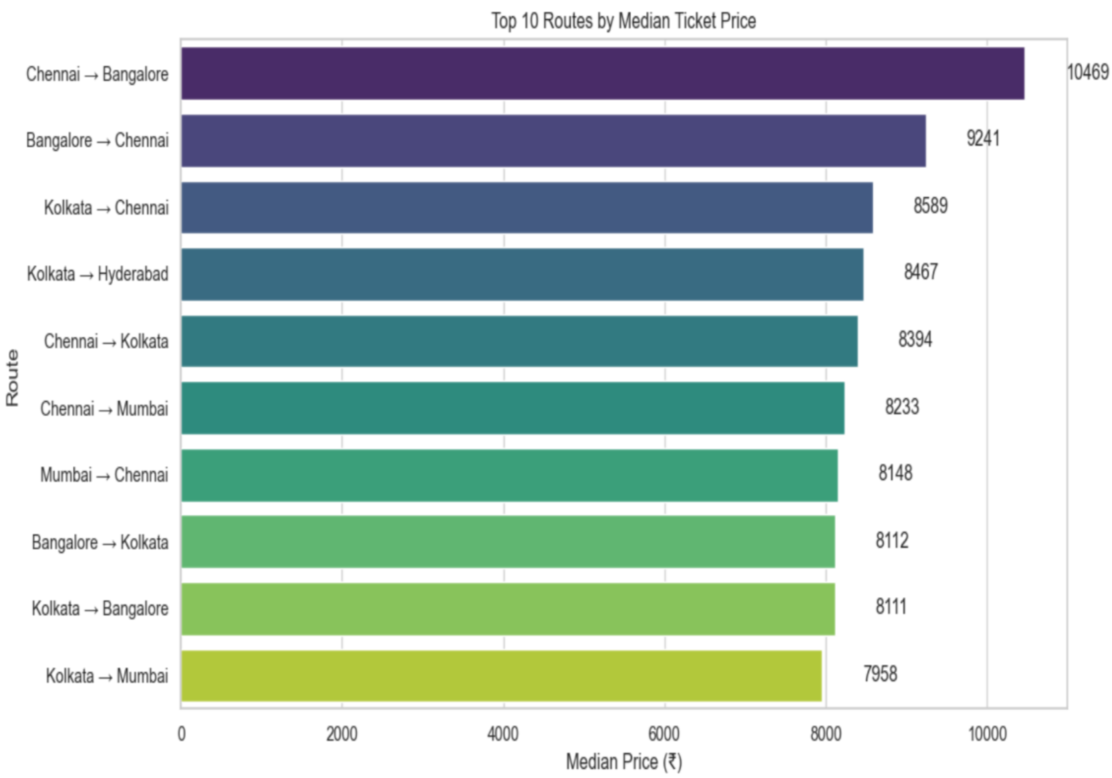


Fig.5.2.2Top ten Routes

Interpretation : Class Distribution (Pie Chart)

The pie chart shows that the majority of airline bookings belong to the Economy class (68.9%), while Business class accounts for only 31.1% of the total bookings. This indicates that most passengers prefer economy tickets due to their affordability and cost-effectiveness.

The lower proportion of Business class suggests that premium travel is limited to a smaller segment of customers, likely due to higher ticket prices. This imbalance highlights that pricing strategies and demand are heavily skewed toward the economy segment.

Interpretation : Top 10 Routes by Median Ticket Price (Bar Chart)

The bar chart reveals that Chennai → Bangalore has the highest median ticket price, followed by Bangalore → Chennai and Kolkata → Chennai, indicating that these routes are relatively more expensive compared to others.

Most of the top routes fall within a similar price range (₹7,900–₹10,500), suggesting moderate variation in median prices across routes. This indicates that while route influences pricing, the variation is not extremely large, and pricing remains somewhat consistent among major city pairs.

Heatmap of Routes

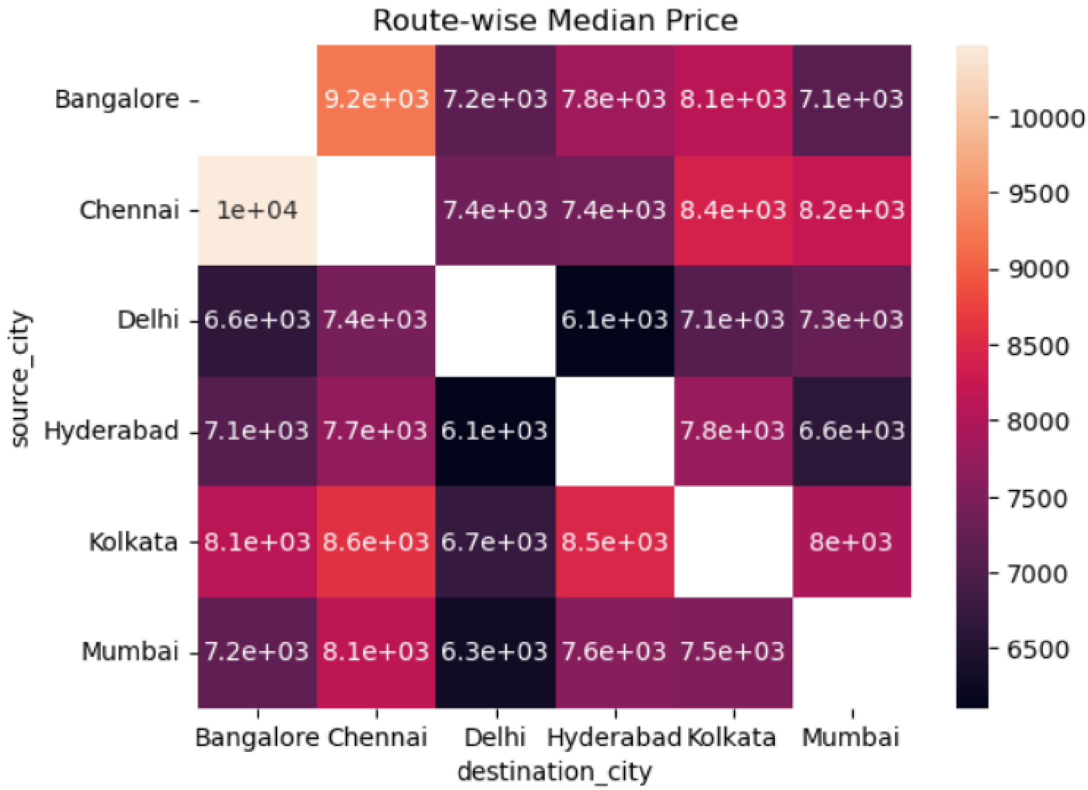


Fig.5.2.3 Heatmap of Source and Destination City

Interpretation:

The heatmap illustrates the variation in median ticket prices across different source and destination city pairs. It is evident that routes such as Chennai → Bangalore and Kolkata → Chennai have relatively higher median prices, indicating strong demand and possibly higher operational or connectivity importance on these routes. In contrast, routes like Delhi → Hyderabad and Mumbai → Delhi show comparatively lower median prices, suggesting more affordable travel options on these routes.

The pricing pattern also reveals that ticket prices are not symmetric across routes, meaning the price from one city to another may differ in the reverse direction. This indicates that factors such as demand imbalance, flight availability, and route popularity influence pricing differently for each direction.

Overall, the heatmap highlights that route plays a significant role in ticket price variation, with certain city pairs consistently showing higher or lower median prices. This supports earlier statistical findings that route is an important determinant of airline ticket pricing.

Conclusion of objective -2

The analysis of airline ticket pricing clearly demonstrates that ticket prices are influenced by multiple key factors, including airline, class, route, stops, duration, and booking timing (days left). Among these, class and airline type have the most significant impact, as Business class tickets and premium airlines such as Vistara and Air India consistently exhibit higher price levels compared to Economy class and low-cost carriers .

Statistical testing further validates these findings. The Kruskal–Wallis test confirms that there are significant differences in ticket prices across both airlines and routes, while the Mann–Whitney U test highlights a strong price difference between Economy and Business classes. These results indicate that pricing is not uniform and varies significantly depending on service quality and travel characteristics.

Additionally, route-level analysis shows that certain high-demand routes, such as Chennai–Bangalore and Bangalore–Chennai, tend to have higher median ticket prices. The distribution of passengers also indicates a strong preference for Economy class, suggesting that affordability plays a major role in customer choice. Overall, the study concludes that airline ticket pricing is a complex, multi-factor system, where both operational and market-driven factors jointly determine price variations, making advanced modelling approaches essential for accurate prediction and decision-making.

5.3 Objective 3: .To examine statistical relationships between variables using techniques like correlation and hypothesis testing.

Spearman Rank Correlation Test

Assumptions

1. Variables should be ordinal, interval, or ratio scale
2. Relationship between variables should be monotonic (increasing or decreasing)
3. Data does not require normal distribution
4. Observations should be independent

Hypothesis

1. Duration vs Price

- Null Hypothesis (H_0):
  There is no significant relationship between duration and ticket price.
- Alternative Hypothesis (H_1):
  There is a significant relationship between duration and ticket price.

2. Days Left vs Price

- Null Hypothesis (H_0):
  There is no significant relationship between days left and ticket price.
- Alternative Hypothesis (H_1):
  There is a significant relationship between days left and ticket price.

Test Results

Variable	Spearman Correlation	p-value
Duration vs Price	0.3188	0.00
Days Left vs Price	-0.2670	0.00

Interpretation

The results show that duration has a positive correlation (0.3188) with ticket price, indicating that longer flights tend to have higher prices. Since the p-value is less than 0.05, this relationship is statistically significant.

On the other hand, days left has a negative correlation (-0.2670) with ticket price, suggesting that as the number of days left before departure increases, ticket prices tend to decrease. This also aligns with real-world scenarios where early booking often results in cheaper fares.

Heatmap All Numeric Variable

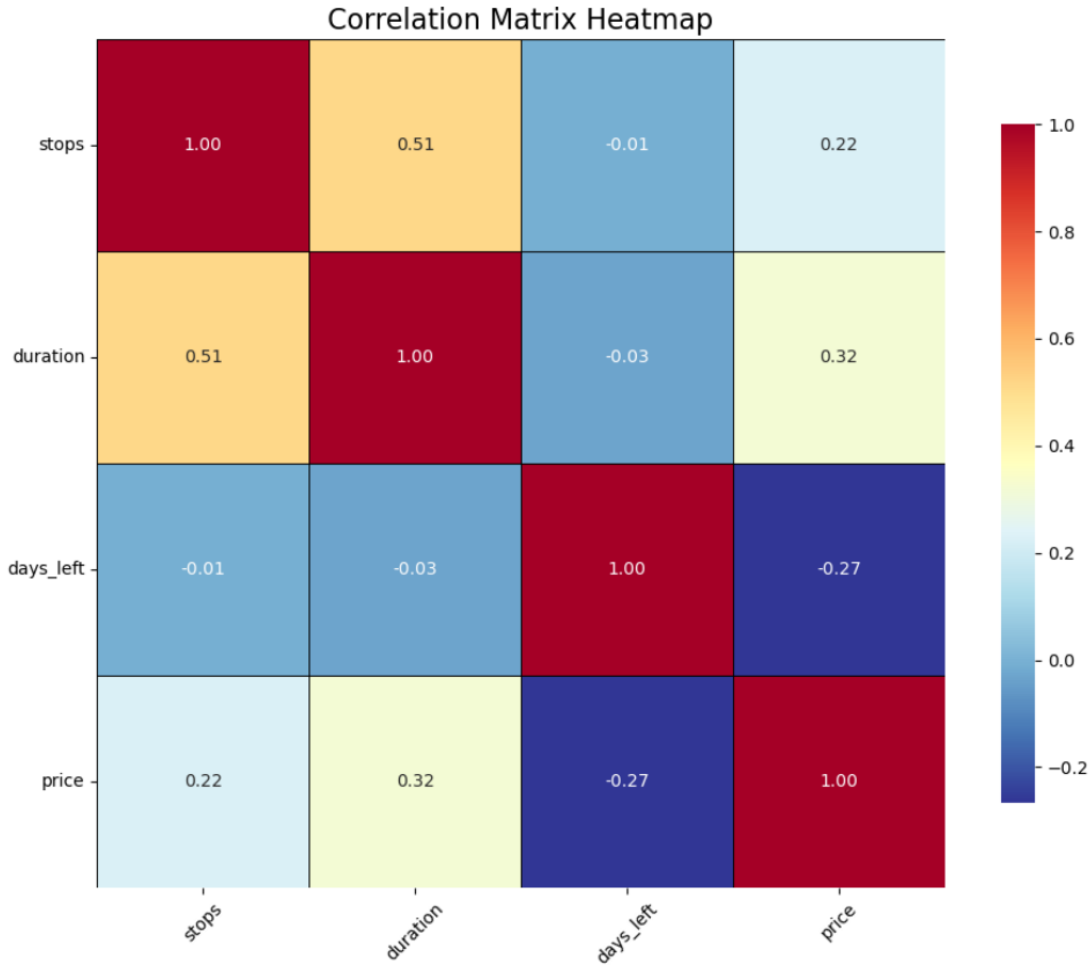


Fig.5.3.1 Heatmap of Numeric Variables

Interpretation:

The correlation heatmap shows the relationships between key numerical variables such as stops, duration, days left, and price. It is observed that duration has a moderate positive correlation with price (0.32), indicating that longer flights tend to have higher ticket prices. Similarly, stops also show a positive correlation with price (0.22), suggesting that flights with more stops are generally associated with higher prices.

A strong positive correlation is observed between stops and duration (0.51), which indicates that flights with more stops usually have longer travel durations. This relationship is logical, as additional stops increase the total journey time.

On the other hand, days left shows a negative correlation with price (-0.27), implying that ticket prices tend to decrease as the number of days left before departure increases. This reflects real-world pricing behavior where early bookings are generally cheaper. Overall, the heatmap confirms that duration, stops, and booking timing are important factors influencing airline ticket prices.

❖ Kruskal–Wallis Test (Stops)

Comparison of Ticket Prices Across Different Numbers of Stops Using Kruskal–Wallis Test

Hypothesis

- Null Hypothesis (H_0):
  There is no significant difference in ticket prices across different numbers of stops.
- Alternative Hypothesis (H_1):
  There is a significant difference in ticket prices across different numbers of stops.

Test Results

- H-statistic: 20,484.30
- p-value: 0.0000041

Median Price by Stops

Stops	Median Price (₹)
2 Stops	8,307
1 Stop	7,959
Non-stop (0)	4,499

Interpretation

Since the p-value is less than the significance level (0.05), the null hypothesis is rejected. This indicates that there is a statistically significant difference in ticket prices across flights with different numbers of stops.

The results show that flights with more stops tend to have higher median ticket prices, while non-stop flights have the lowest median prices. This suggests that the number of stops is an important factor influencing ticket pricing, possibly due to differences in route complexity, travel duration, and demand patterns.

❖ Mann–Whitney U Test (Direct Flights vs Connecting Flights)

Comparison of Ticket Prices Between Non-Stop Flights and Flights with Stops Using Mann–Whitney U Test

Mann–Whitney U Test

Hypothesis

- Null Hypothesis (H_0):
  There is no significant difference in ticket prices between non-stop flights (0 stops) and flights with stops (>0).
- Alternative Hypothesis (H_1):
  There is a significant difference in ticket prices between non-stop flights and flights with stops.

Test Results

- U-statistic: 2,549,486.50
- p-value: 0.000

Median Price Comparison

Flight Type	Median Price (₹)
Non-stop (0 Stops)	4,499
With Stops (>0 Stops)	8,016

Interpretation

Since the p-value is less than the significance level (0.05), the null hypothesis is rejected. This indicates that there is a statistically significant difference in ticket prices between non-stop flights and flights with stops.

The results show that flights with stops have higher median ticket prices compared to non-stop flights. This suggests that the presence of stops significantly influences pricing, possibly due to longer travel duration, route complexity, or demand patterns.

Day left vs Ticket Price Line chart

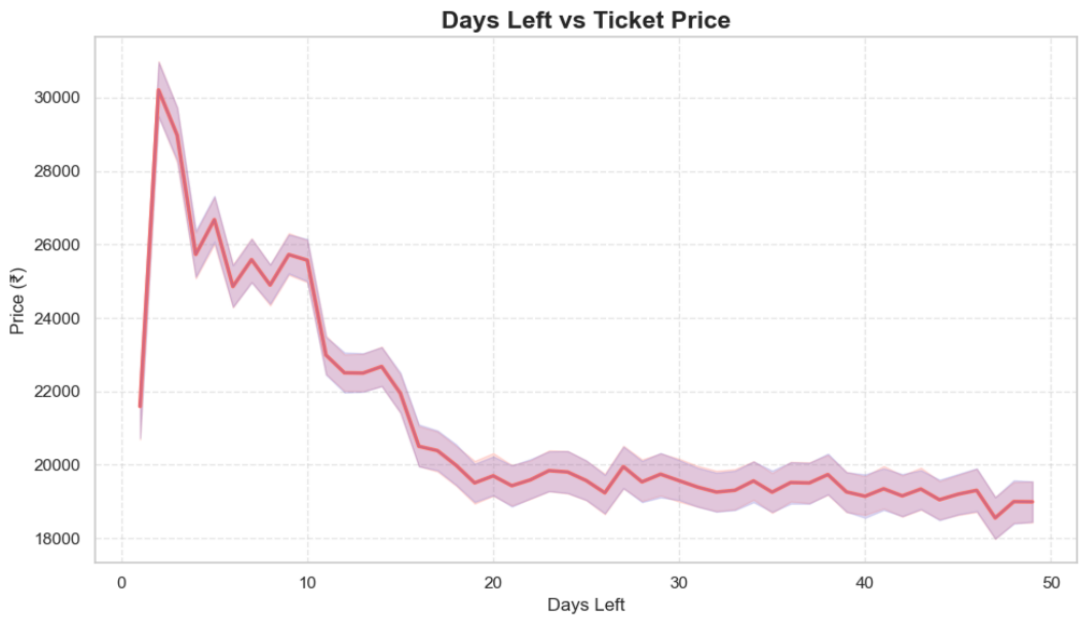


Fig.5.3.2 day Left vs ticket Price Line Chart

Interpretation:

The plot shows a clear downward trend in ticket prices as the number of days left before departure increases. Prices are highest when the booking is made very close to the departure date (around 1–5 days), indicating last-minute bookings are significantly more expensive. This reflects high demand and urgency pricing by airlines.

As the number of days left increases, ticket prices gradually decrease and stabilize after around 20 days. This suggests that early booking leads to lower and more stable prices, making it more economical for passengers to plan their travel in advance.

Overall, the chart highlights a negative relationship between days left and ticket price, confirming that booking timing is a crucial factor in airline pricing strategies, where earlier bookings generally result in cheaper fares.

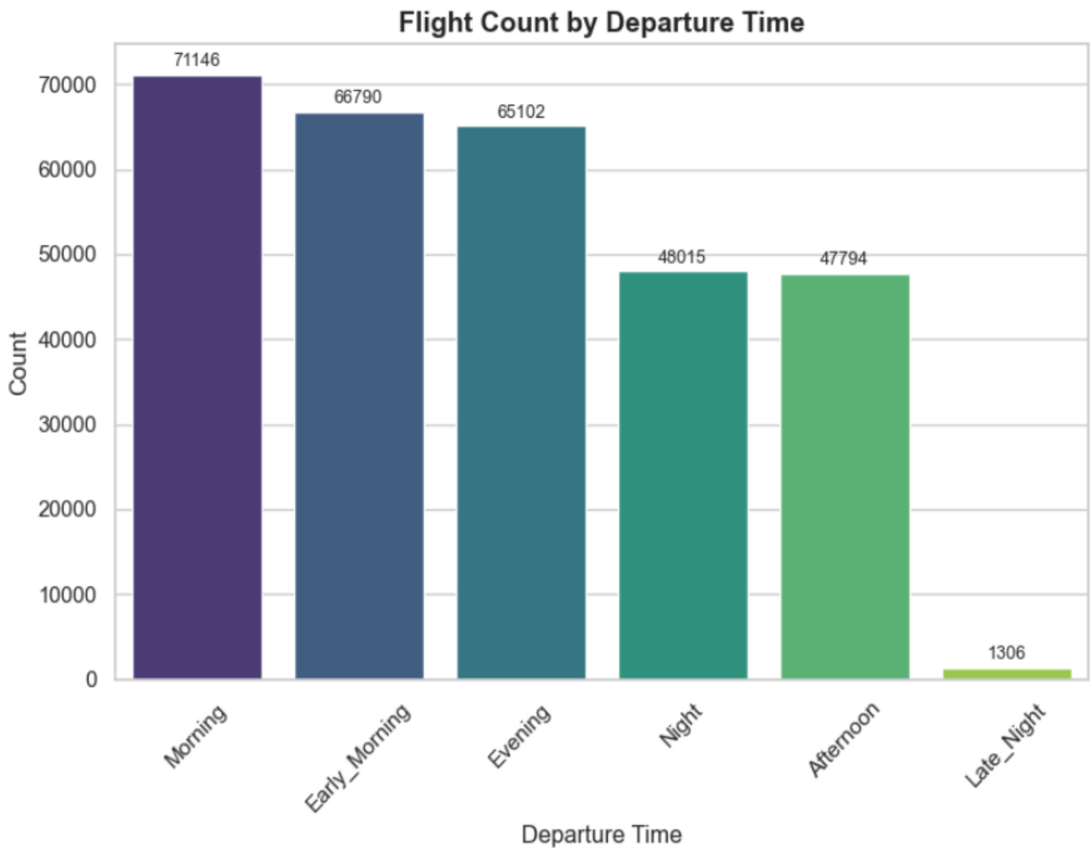


Fig.5.3.3 Departure time Count Plot

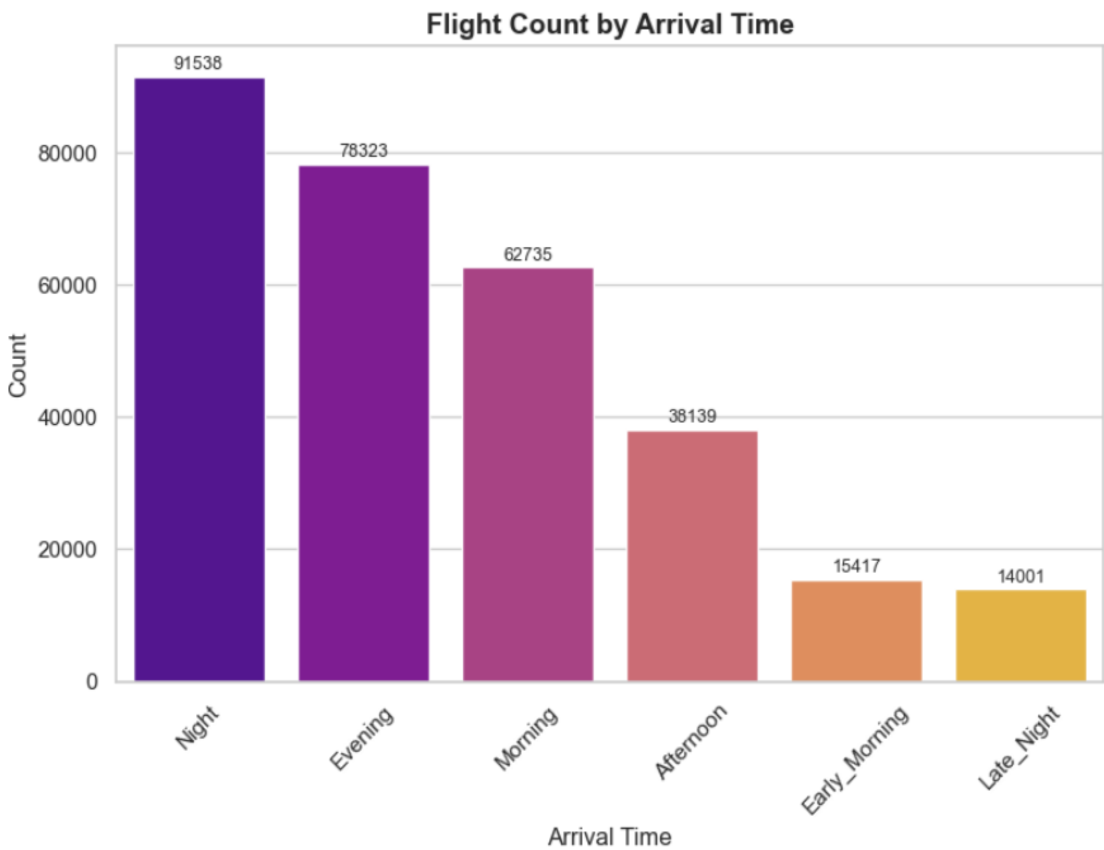


Fig.5.3.4 Arrival Time count plot

Interpretation: Flight Count by Departure Time

The chart shows that most flights depart during Morning (71,146), Early Morning (66,790), and Evening (65,102), indicating that these time slots are the busiest for departures. This suggests that airlines schedule more flights during these periods to match high passenger demand, especially for business and daily commuters.

In contrast, Night and Afternoon departures have moderate frequencies, while Late Night has the lowest number of flights (1,306). This indicates that late-night travel is less preferred, possibly due to inconvenience and lower passenger demand.

Interpretation: Flight Count by Arrival Time

The arrival time distribution shows that Night arrivals are the highest (91,538), followed by Evening (78,323) and Morning (62,735). This suggests that many flights are scheduled to arrive during night hours, likely due to longer travel durations or scheduling efficiency.

On the other hand, Early Morning and Late Night arrivals have the lowest counts, indicating fewer flights are scheduled during these less convenient hours. This pattern reflects airline strategies to align arrival times with passenger convenience and airport operations.

Fig.5.3.5 Booking Group By Median Price Bar chart

Interpretation:

The chart shows that ticket prices are highest for last-minute bookings (0–5 days), with a median price of around ₹14,744. Prices gradually decrease as the number of days before departure increases, with 6–15 days also relatively expensive (₹12,062). This indicates that booking closer to the travel date results in significantly higher fares.

For bookings made earlier, such as 16–30 days (₹6,228) and 31–60 days (₹6,015), the median prices are much lower and relatively stable. This highlights that early booking leads to more affordable ticket prices, making it beneficial for passengers to plan in advance.

Additionally, the number of observations (n) increases for longer booking periods, suggesting that most passengers prefer booking tickets earlier. Overall, the chart clearly demonstrates that booking timing is a critical factor affecting airline ticket pricing, with early bookings offering cost advantages.

Conclusion for Objective 3

The statistical analysis confirms that airline ticket prices are significantly influenced by multiple variables, particularly duration, number of stops, and booking timing (days left). The Spearman correlation results indicate that duration has a positive relationship with price, meaning longer flights generally cost more, while days left shows a negative relationship, confirming that early bookings are associated with lower ticket prices. These relationships are statistically significant, highlighting their importance in understanding pricing behaviour.

Further, hypothesis testing using the Kruskal–Wallis test and Mann–Whitney U test demonstrates that ticket prices differ significantly across flight categories, especially based on the number of stops and type of journey (non-stop vs connecting). Flights with stops tend to have higher median prices compared to non-stop flights, indicating that route complexity and travel structure play a crucial role in pricing. These results confirm that price variation is not random but statistically driven by key travel characteristics.

Additionally, visual analyses such as heatmaps and trend plots reinforce these findings by showing consistent patterns across variables. The clear downward trend between days left and price highlights the importance of booking time, while the relationships among stops, duration, and price indicate interconnected effects. Overall, the objective concludes that airline ticket pricing is governed by significant and measurable statistical relationships, making these variables essential for predictive modelling and pricing strategy analysis.

5.4 Objective 4: To apply quantile regression to study how different factors influence low, median, and high ticket prices.

Quantile Regression Model

Assumptions of Quantile Regression

1. Linearity in Parameters
2. Independence of Observations
3. No Perfect Multicollinearity
4. Correct Model Specification
5. Existence of Conditional Quantiles
6. Large Sample Size
7. No Quantile Crossing

Quantile Regression Coefficients (25th Quantile – Low Price Segment)

Variable	Coefficient	Std. Error	t-value	p-value
Constant	41,920.00	34.881	1201.847	0.000
Airline	413.79	3.484	118.779	0.000
Source City	131.85	3.572	36.912	0.000
Departure Time	10.48	3.526	2.973	0.003
Stops	2239.33	18.576	120.547	0.000
Arrival Time	35.36	3.532	10.009	0.000
Destination City	142.39	3.561	39.985	0.000
Class	-40,030.00	13.855	-2889.504	0.000
Duration	1.23	0.017	71.494	0.000
Days Left	-81.47	0.397	-205.052	0.000

Quantile Regression Coefficients (50th Quantile – Median Price Segment)

Variable	Coefficient	Std. Error	t-value	p-value
Constant	50,770.00	35.178	1443.126	0.000
Airline	516.59	3.657	141.259	0.000
Source City	35.83	3.854	9.298	0.000
Departure Time	17.24	3.784	4.556	0.000
Stops	2312.19	18.798	123.005	0.000
Arrival Time	52.74	3.787	13.927	0.000
Destination City	52.37	3.869	13.534	0.000
Class	-46,500.00	14.620	-3180.408	0.000
Duration	1.16	0.018	65.875	0.000
Days Left	-113.51	0.485	-234.033	0.000

Quantile Regression Coefficients (75th Quantile – High Price Segment)

Variable	Coefficient	Std. Error	t-value	p-value
Constant	57,430.00	43.004	1335.446	0.000
Airline	575.75	4.518	127.439	0.000
Source City	54.37	4.966	10.949	0.000
Departure Time	34.28	4.777	7.176	0.000
Stops	2775.99	24.378	113.872	0.000
Arrival Time	57.03	4.789	11.910	0.000
Destination City	69.46	5.014	13.853	0.000
Class	-51,180.00	17.972	-2847.594	0.000
Duration	1.04	0.022	46.957	0.000
Days Left	-146.99	0.695	-211.518	0.000

Interpretation

The quantile regression model explains how different factors affect airline ticket prices across low (25th), median (50th), and high (75th) price levels. The results show that the impact of variables is not constant but varies across different price segments, making this model more suitable than traditional regression. It effectively captures the heterogeneous nature of ticket pricing in the dataset.

The class variable has the strongest influence on price, with a large negative coefficient across all quantiles, and its effect increases at higher price levels. This indicates a significant price difference between economy and business class, especially for expensive tickets. Similarly, the number of stops has a strong positive impact, which becomes larger at higher quantiles, showing that flights with more stops tend to be more expensive.

The variable days_left shows a consistent negative relationship with price, and its effect becomes stronger from lower to higher quantiles. This confirms that ticket prices decrease as the number of days before departure increases, meaning last-minute bookings are costlier. Additionally, the airline variable has a positive and increasing effect, indicating that premium airlines contribute more to higher ticket prices.

Other variables such as departure time, arrival time, and destination city have moderate positive effects, while duration has a small but consistent impact on price. Overall, the model is stable and all variables are significant, showing that quantile regression successfully captures variations in pricing behaviour across different ticket price levels and is well-suited for this dataset.

Quantile Regression Coefficients (25th, 50th, 75th Quantiles)

Variable	25th Quantile	50th Quantile (Median)	75th Quantile
Constant	41922.07	50765.73	57429.40
Airline	413.79	516.59	575.75
Source City	131.85	35.83	54.37
Departure Time	10.48	17.24	34.28
Stops	2239.33	2312.19	2775.99
Arrival Time	35.36	52.74	57.03
Destination City	142.39	52.37	69.46
Class	-40034.91	-46497.06	-51176.36
Duration	1.23	1.16	1.04
Days Left	-81.47	-113.51	-146.99

Interpretation :

The quantile regression results show how different factors influence airline ticket prices across low (25th), median (50th), and high (75th) price levels. It is observed that variables such as airline, stops, and class have a strong and consistent impact across all quantiles. In particular, the coefficient for stops increases from 2239 (25th) to 2775 (75th), indicating that additional stops significantly increase ticket prices, especially for higher-priced tickets. Similarly, the airline effect becomes stronger at higher quantiles, suggesting premium airlines charge more in the higher price segment.

The class variable has a large negative coefficient across all quantiles, with its magnitude increasing from -40,034 to -51,176. This indicates a strong difference between economy and business class pricing, and the impact becomes more pronounced at higher price levels. Additionally, days_left shows a negative effect, which becomes stronger from -81 to -146, confirming that ticket prices decrease as the number of days before departure increases, especially for expensive tickets.

Other variables such as departure time, arrival time, and destination city show moderate positive effects, with slightly higher influence at upper quantiles. In contrast, duration has a small but consistent positive effect, indicating longer flights are slightly more expensive. Overall, the results highlight that pricing factors behave differently across price levels, making quantile regression a suitable model for capturing these variations compared to traditional regression methods.

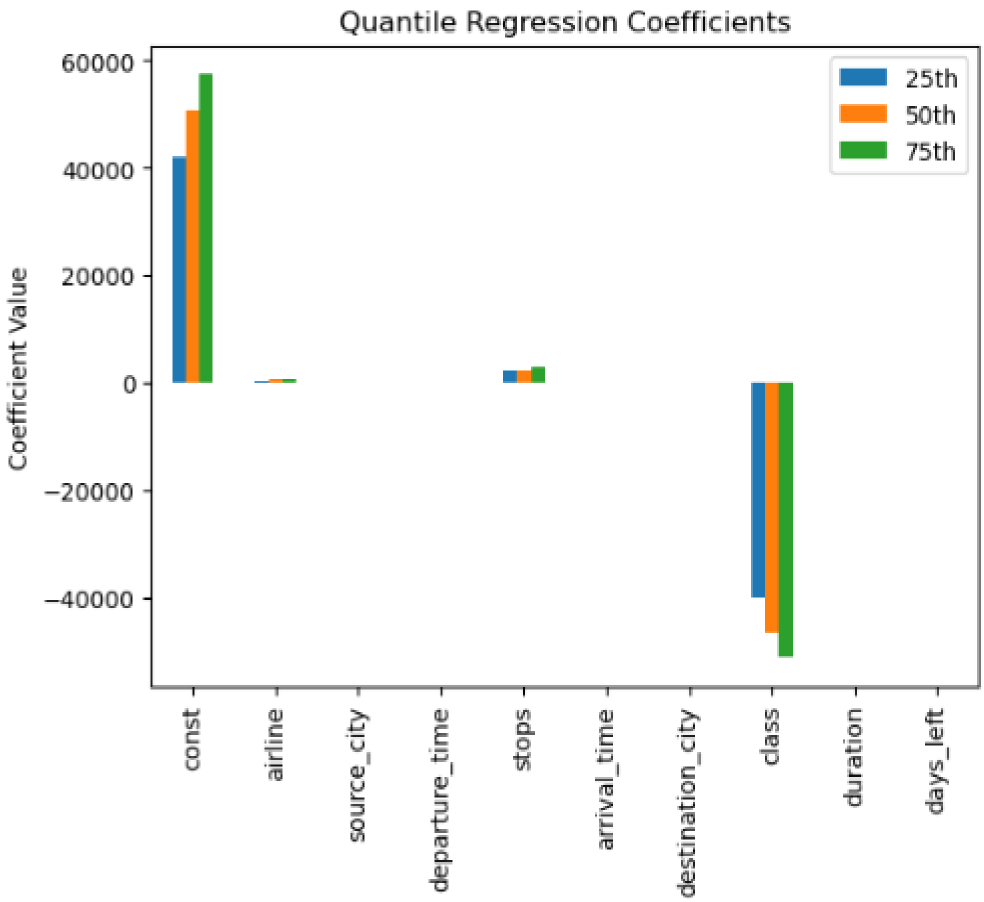


Fig.5.4.1 Quantile Regression Coefficients Charts

Interpretation:

The chart shows that the impact of variables on ticket price changes across different quantiles (low, median, high prices). Most variables have consistent effects, but their strength increases at higher price levels.

Class has the strongest negative impact, especially for high-priced tickets, indicating a major difference between economy and business pricing. Stops and airline show increasing positive effects, meaning they influence high prices more.

Days_left has a negative effect that becomes stronger at higher quantiles, confirming that last-minute bookings are more expensive. Overall, the model captures how pricing factors vary across different price segments.

Pseudo R^2 (Median): 0.737298898121596

MAE: 4242.146999076945

Train-test split is used to evaluate the model's performance on unseen data and ensure it generalizes well without overfitting**.**

After train Model:

Test MAE: 4221.25171369128

Interpretation

The model shows a Pseudo R^2 value of 0.7373 (median model), indicating that approximately 73.7% of the variation in ticket prices is explained by the model. This suggests a strong model fit, especially considering the complex and skewed nature of airline pricing data.

The Mean Absolute Error (MAE) is around ₹4224 (training) and ₹4221 (test), which indicates that the model's predictions are, on average, off by about ₹4200. The similarity between training and testing MAE values shows that the model is well-generalized and does not suffer from overfitting.

Additionally, the condition checks show that pred_25 ≤ pred_50 ≤ pred_75 (True), confirming that the quantile predictions are logically consistent and there is no quantile crossing issue, which validates the correctness of the model.

Overall, the results indicate that the quantile regression model is reliable, stable, and appropriate for this dataset. It successfully captures variations in ticket prices across different segments and provides accurate predictions, making it suitable for analyzing and forecasting airline ticket pricing behaviour.

```python
residuals = y_test - y_pred
```

```python
residuals
```

```
27131      3033.583944
266857    12477.585274
141228      479.864658
288329     4627.870368
97334       650.970377
              ...
5234       -340.012030
5591       -311.709710
168314     1498.177675
175191     3204.642349
287693     8880.623907
Length: 60031, dtype: float64
```

Interpretation:

The residuals represent the difference between actual and predicted ticket prices. The values include both positive and negative residuals, indicating that the model sometimes underestimates (positive residuals) and sometimes overestimates (negative residuals) the ticket prices.

The presence of both small and large residual values shows that while many predictions are close to actual values, some observations have higher prediction errors, especially for extreme prices. This suggests that the model performs reasonably well overall but may have difficulty accurately predicting certain high or complex cases.

Overall, the residuals indicate that the model is fairly balanced with no strong bias, but the variability in residuals suggests there is still some unexplained variation in ticket prices, likely due to factors not included in the model.

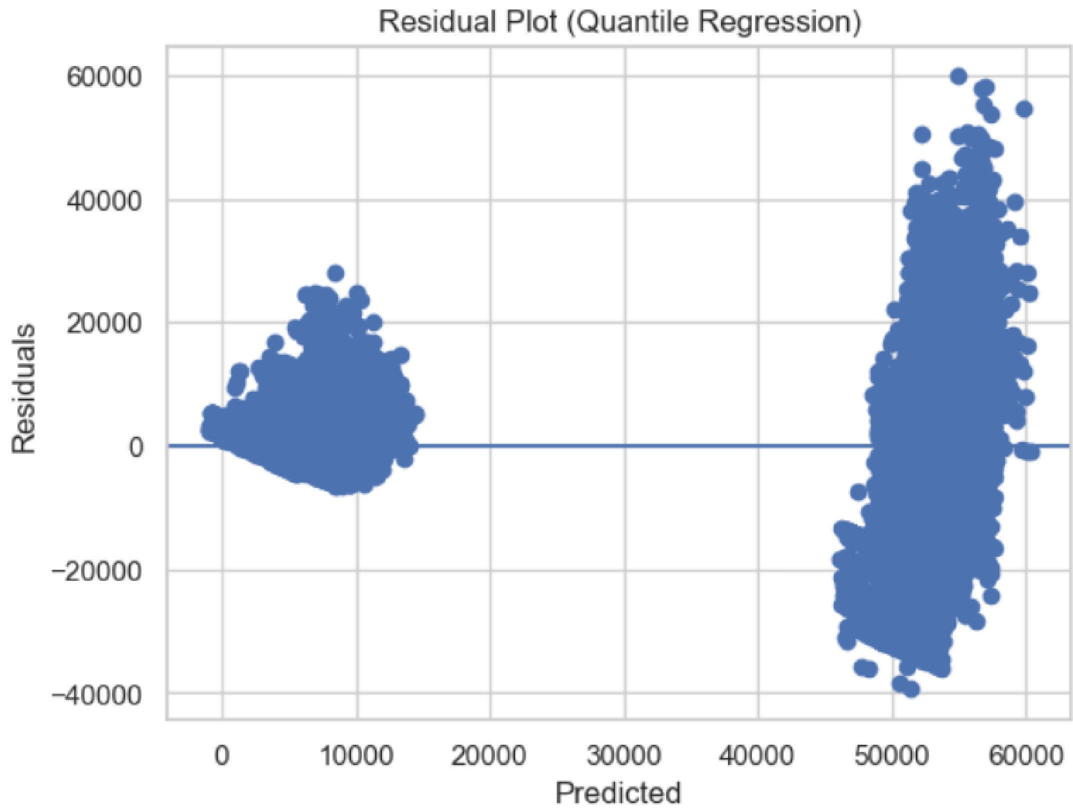


Fig.5.4.2 Residuals Plots

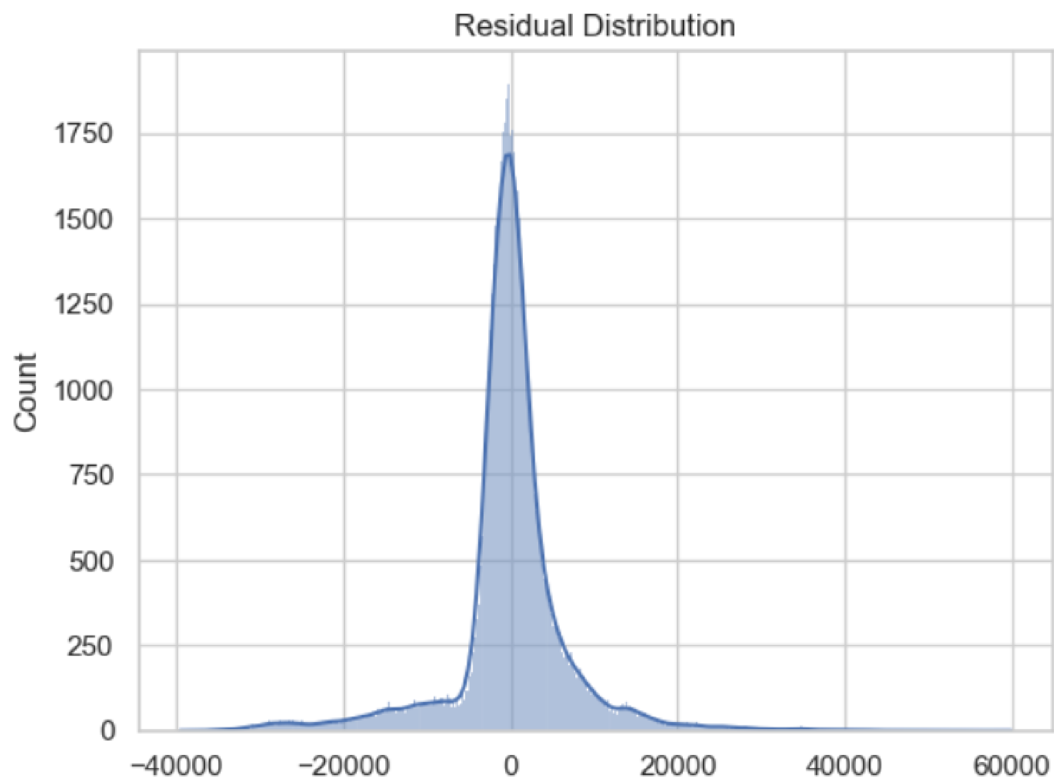


Fig.5.4.3 Residuals Distribution Plots

Interpretation

The residual plot shows that residuals are scattered around zero, indicating that the model does not have a strong systematic bias. However, the spread of residuals increases for higher predicted values, forming a wider pattern. This suggests the presence of heteroscedasticity, meaning the model has higher prediction errors for expensive tickets compared to cheaper ones.

The residual distribution plot shows that most residuals are concentrated around zero, indicating that the model performs well for a large number of observations. However, the distribution has long tails on both sides, showing the presence of large errors and outliers, especially for extreme price values.

Overall, the model provides reasonably accurate predictions, but the variability in residuals and presence of heteroscedasticity indicate that prediction accuracy decreases for high-priced tickets. This suggests that while the model is effective, there is still some unexplained variation in the data.

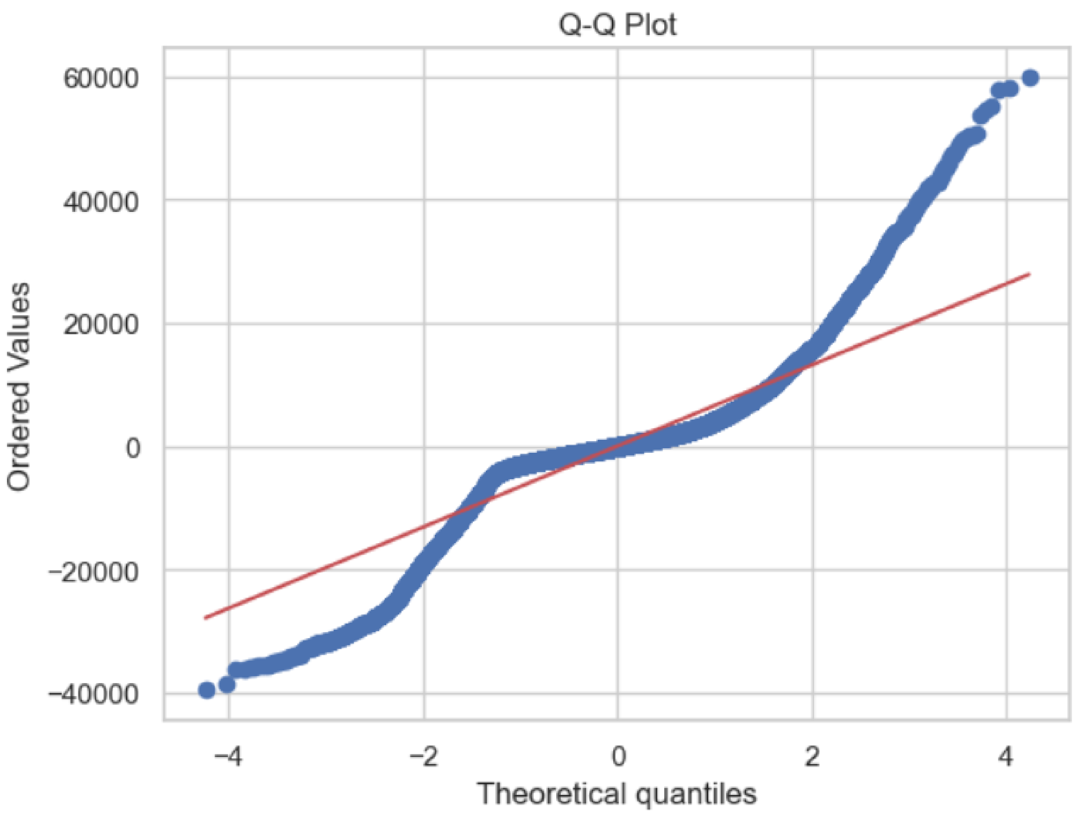


Fig.5.4.4 Q-Q Plot

Interpretation:

The Q-Q plot shows that residuals are not normally distributed, as points deviate from the straight line. This indicates skewness and presence of extreme values. Overall, it supports using quantile regression since it does not require normality.

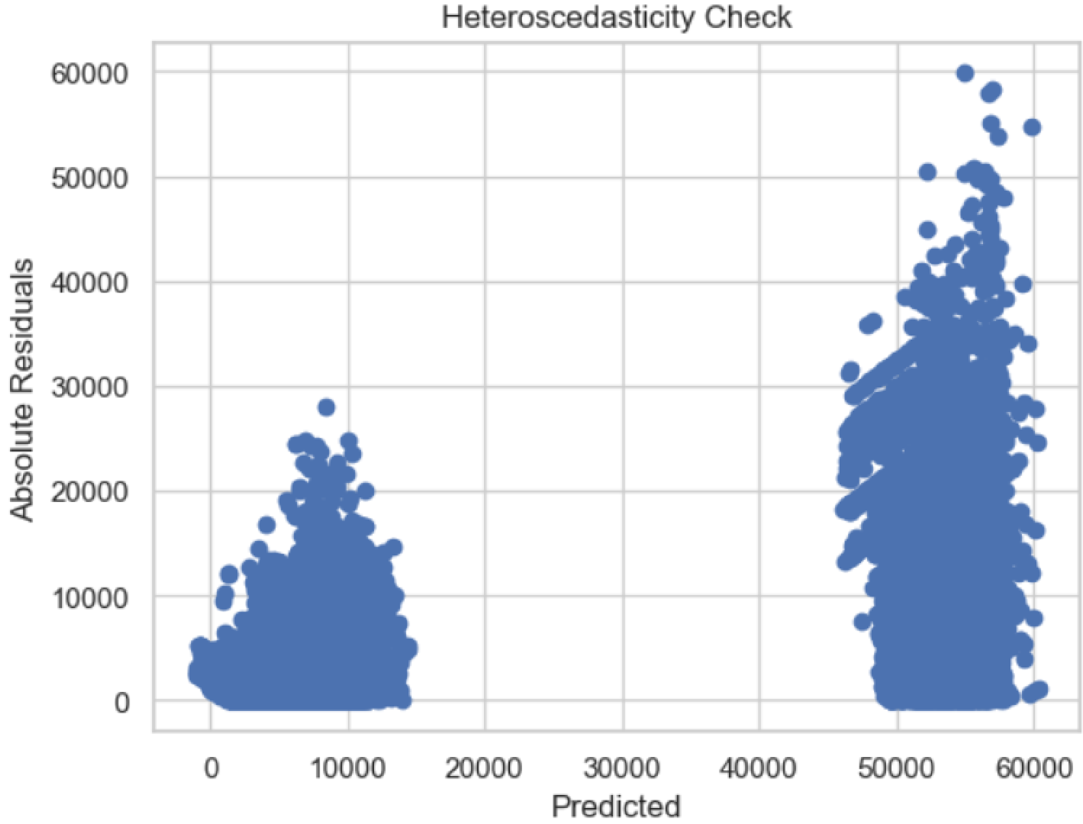


Fig.5.4.5 Heteroscedasticity check plot

Interpretation:

The mean residual is approximately 54.88, which is close to zero, indicating that the model does not have a strong overall bias and predictions are generally balanced. However, the heteroscedasticity plot shows that the spread of residuals increases as the predicted values increase, meaning that the variability of errors is higher for larger predicted prices. This indicates the presence of heteroscedasticity, where the model's prediction errors are not constant across all levels of price. Overall, the model performs well for lower-priced tickets but becomes less accurate for high-priced tickets due to increased variability in residuals.

Conclusion of Objective 4

The quantile regression analysis successfully achieved the objective of examining how different factors influence airline ticket prices across low, median, and high price segments. The results clearly show that the effect of variables is not constant but varies across different quantiles, highlighting the heterogeneous nature of pricing behaviour. Key variables such as class, stops, airline, and days_left have a strong and significant impact on ticket prices, with their influence becoming more pronounced in higher price segments.

The model demonstrates strong performance with a Pseudo R^2 of approximately 0.73 and consistent MAE values (~₹4200) for both training and testing datasets, indicating good predictive accuracy and no overfitting. Additionally, the condition ($pred\_25 \leq pred\_50 \leq pred\_75$) confirms that the model is logically consistent and satisfies the non-crossing quantile assumption, validating its correctness.

Diagnostic analysis shows that residuals are centered around zero, indicating no major bias, although the presence of heteroscedasticity suggests higher prediction errors for expensive tickets. The Q-Q plot further confirms that residuals are not normally distributed, which supports the use of quantile regression as it does not require normality assumptions.

Overall, the quantile regression model is robust, reliable, and well-suited for this dataset. It effectively captures variations in ticket pricing across different segments and provides deeper insights compared to traditional regression, making it highly useful for understanding and predicting airline ticket prices.

5.5 Objective 5: To develop a Gamma regression model for accurate prediction of airline ticket prices considering skewed data.

Generalized Linear Model Regression Results

Dep. Variable:	price	No. Observations:	235966
Model:	GLM	Df Residuals:	235956
Model Family:	Gamma	Df Model:	9
Link Function:	Log	Scale:	0.13037
Method:	IRLS	Log-Likelihood:	-2.2823e+06
Date:	Thu, 02 Apr 2026	Deviance:	27341.
Time:	17:48:42	Pearson chi2:	3.08e+04
No. Iterations:	18	Pseudo R-squ. (CS):	0.9998
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	10.2905	0.004	2362.612	0.000	10.282	10.299
duration	7.464e-05	2.17e-06	34.400	0.000	7.04e-05	7.89e-05
days_left	-0.0151	5.5e-05	-274.888	0.000	-0.015	-0.015
stops	0.3850	0.002	178.402	0.000	0.381	0.389
class_Economy	-2.0086	0.002	-1111.914	0.000	-2.012	-2.005
airline_Air_India	0.4658	0.004	126.568	0.000	0.459	0.473
airline_GO_FIRST	0.3775	0.004	90.698	0.000	0.369	0.386
airline_Indigo	0.3135	0.004	82.911	0.000	0.306	0.321
airline_SpiceJet	0.4459	0.005	81.592	0.000	0.435	0.457
airline_Vistara	0.5825	0.004	164.219	0.000	0.576	0.589

Gamma Model:

MAE: 4691.027877838277

RMSE: 7881.778489854462

Pseudo R²: 0.9040849612592035

Interpretation

The Gamma GLM model shows strong predictive performance, with a Pseudo R^2 of 0.904, indicating that the model explains about 90.4% of the variation in ticket prices. The MAE (₹4691) and RMSE (₹7881) values suggest that the prediction error is moderate and acceptable given the wide range of ticket prices. Overall, the model provides a good fit for the data.

Coefficient Interpretation

- Duration : A positive coefficient indicates that longer flights lead to higher ticket prices.
- Days Left : A negative coefficient shows that prices decrease as the number of days before departure increases, confirming that early booking is cheaper.
- Stops : Flights with more stops tend to have higher prices, likely due to longer travel routes and added complexity.
- Class : The strong negative coefficient for Economy class indicates that Economy tickets are significantly cheaper compared to Business class.

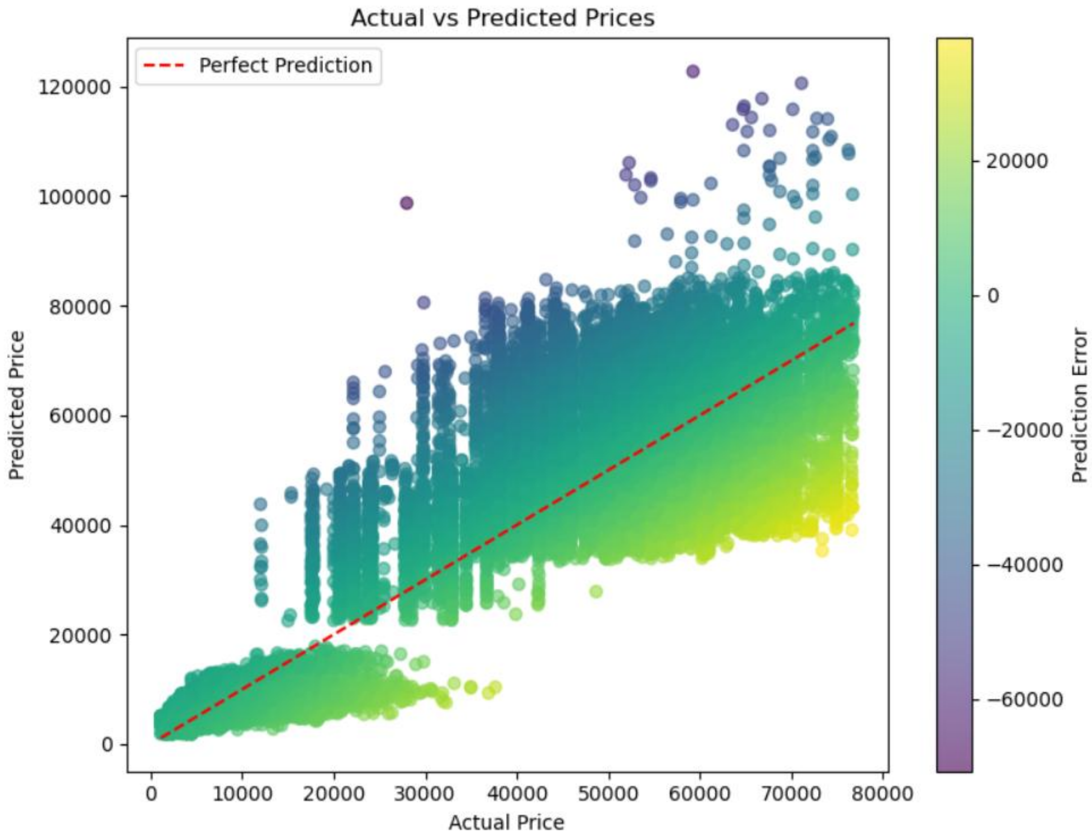


Fig.5.5.1 Actual v/s Predicted Price Plot

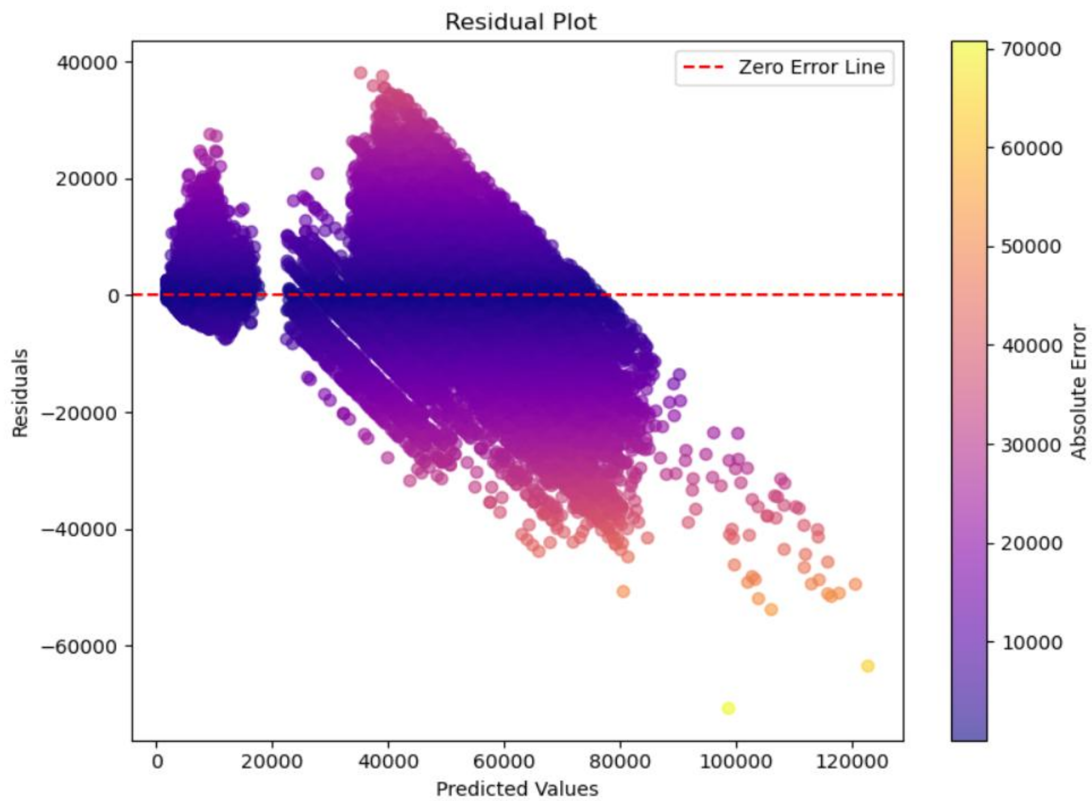


Fig.5.5.2 Residuals Plots

Interpretation:

Actual vs Predicted Prices Plot

The Actual vs Predicted Prices plot shows how closely the model's predictions match the true ticket prices. Ideally, all points should lie along the diagonal "perfect prediction" line, but the scatter indicates deviations, especially at higher price levels. The model performs relatively well for lower and mid-range prices, where points are closer to the line. However, as ticket prices increase, the spread becomes wider, indicating larger prediction errors. This suggests that the model struggles to accurately predict expensive tickets and tends to underestimate high values while slightly overestimating lower ones.

Residual Plot

The residual plot displays the difference between actual and predicted values against predicted prices. Instead of random scatter, a clear funnel-shaped pattern is observed, indicating heteroscedasticity (non-constant variance). As predicted values increase, the spread of residuals also increases, meaning the model's error grows for higher-priced tickets. Additionally, the downward trend suggests a systematic bias, where the model underestimates higher prices and overestimates lower prices. This pattern indicates that while the model captures general trends, it does not fully account for variability in high-value observations.

	Feature	VIF
0	duration	6.190104
1	days_left	4.123453
2	stops	7.598208
3	class_Economy	3.518194
4	airline_Air_India	3.269638
5	airline_GO_FIRST	1.618018
6	airline_Indigo	2.007835
7	airline_SpiceJet	1.238410
8	airline_Vistara	4.133540

Durbin-Watson: 1.9931845521939036

	Actual	Predicted
133477	6909	6880.489089
78972	9201	7900.583604
138000	24686	11927.806628
17058	6832	5144.267323
53979	5102	7856.200908

	Model	MAE	RMSE	R2
0	Gamma GLM	4691.027878	7881.77849	0.904085

Interpretation:

Multicollinearity Check (VIF)

The Variance Inflation Factor (VIF) values for all variables are below 10, indicating that there is no serious multicollinearity among the independent variables. Most variables have VIF values between 1 and 5, suggesting only moderate correlation, which is acceptable. The highest VIF is for stops (~7.59), but it is still within the acceptable range. Therefore, the predictors are not highly correlated, and the model estimates are reliable.

Durbin–Watson Test

The Durbin–Watson statistic is approximately 1.99, which is very close to 2. This indicates that there is no autocorrelation in residuals, meaning the errors are independent. This satisfies one of the key assumptions of regression models and confirms that the model does not suffer from serial correlation issues.

Actual vs Predicted Values

The comparison between actual and predicted values shows that the model is able to capture the general trend of ticket prices. However, there are noticeable differences between actual and predicted values in some cases, indicating prediction errors. The model performs better for lower and moderate prices, while deviations are larger for higher prices, suggesting some limitations in predicting extreme values.

Model Performance Metrics

The model evaluation metrics indicate good performance. The MAE (~₹4691) shows that, on average, predictions deviate from actual values by around ₹4691, which is reasonable given the wide price range. The RMSE (~₹7881) indicates moderate variability in prediction errors. The Pseudo R^2 (~0.904) suggests that the model explains about 90.4% of the variation in ticket prices, indicating a strong fit.

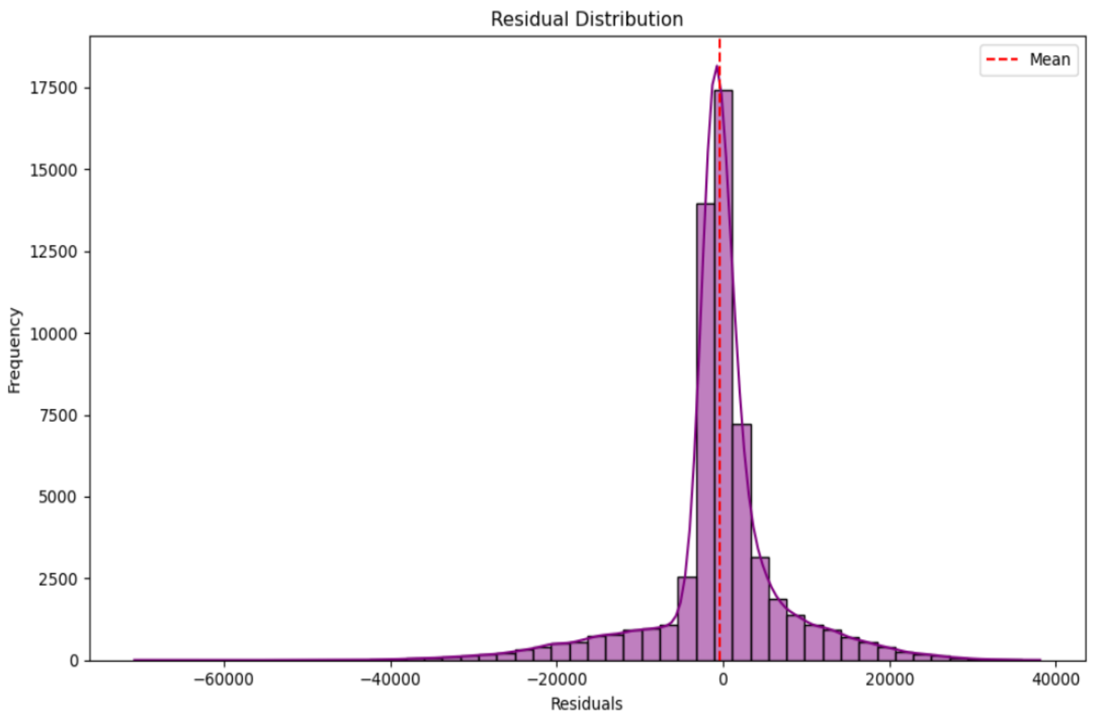


Fig.5.5.3 Residuals Distribution Plot

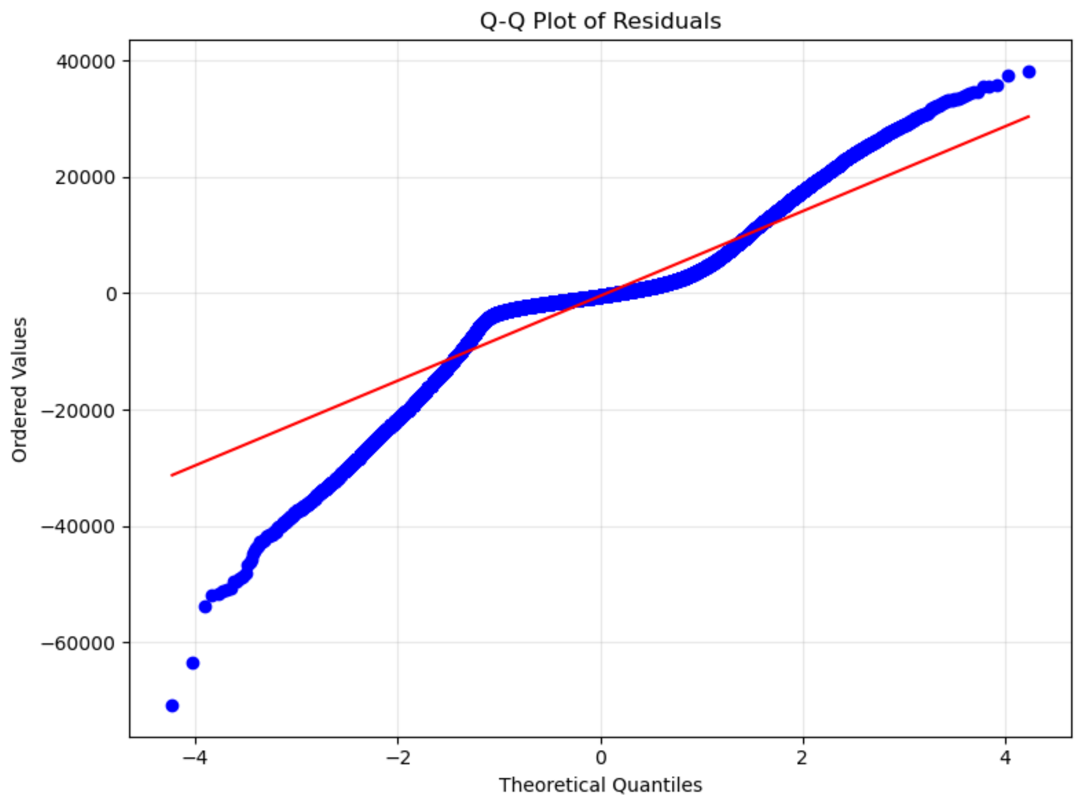


Fig.5.5.4 QQ Plots of Residuals

Interpretation:

Q–Q Plot Interpretation

The Q–Q plot compares the distribution of residuals with a normal distribution. Ideally, if residuals are normally distributed, the points should lie along the straight reference line. However, in this plot, the points show a clear S-shaped deviation from the line, especially at both tails. The lower tail deviates downward while the upper tail deviates upward, indicating the presence of heavy tails and non-normality. This suggests that extreme values (both high and low residuals) are more frequent than expected under normal distribution. Such behaviour is common in skewed data like airline prices and indicates that the residuals are not perfectly normally distributed.

Residual Distribution Interpretation

The residual distribution plot shows that most residuals are concentrated around zero, forming a peak near the center , which indicates that the model predictions are generally unbiased. However, the distribution is slightly skewed and has long tails, particularly on the negative side, indicating the presence of large prediction errors. The spread of residuals on both sides suggests variability in prediction accuracy, especially for extreme values. The deviation from a symmetric bell-shaped curve confirms that the residuals do not follow a perfect normal distribution.

Conclusion of Objective 5

The Gamma regression model was successfully developed to predict airline ticket prices, effectively handling the skewed and non-negative nature of the data. The model shows strong performance with a high Pseudo R^2 value (≈ 0.904), indicating that it explains a large proportion of variation in ticket prices. Additionally, the error metrics (MAE ≈ ₹4691 and RMSE ≈ ₹7881) suggest that the model provides reasonably accurate predictions across most observations.

The model results highlight that important variables such as duration, number of stops, days left before departure, class, and airline significantly influence ticket prices. The model captures real-world trends effectively, such as higher prices for longer flights and flights with more stops, and lower prices for early bookings and economy class. This confirms that both operational and market-related factors play a key role in determining airline ticket prices.

Diagnostic checks confirm that the model satisfies key assumptions, with no serious multicollinearity (VIF < 10) and no autocorrelation (Durbin–Watson ≈ 2). Although residual analysis indicates heteroscedasticity and non-normality, these are expected in skewed datasets and are appropriately handled by the Gamma GLM. Overall, the model is robust, reliable, and well-suited for predicting airline ticket prices.

5.6 Objective 6 :To build and compare machine learning models to improve prediction accuracy.

1. Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical outcome. It assumes that the target can be represented through a linear relationship with the input information.

In this project, Linear Regression was used as a baseline model for ticket price prediction. It provides a simple reference point against which the performance of more complex machine learning algorithms can be compared.

Key idea: It estimates the best-fitting linear relationship between the predictors and the target by minimizing prediction errors.

Purpose in the project:
To establish a baseline prediction performance and understand how well a simple linear model can predict airline ticket prices.

---

2. Ridge Regression

Ridge Regression is an extension of Linear Regression that incorporates L2 regularization. The regularization term penalizes large model coefficients, helping to reduce overfitting and improve model stability.

Ridge Regression is particularly useful when the dataset contains several related or correlated predictors.

In this project, Ridge Regression was used as a regularized linear model and compared with ordinary Linear Regression.

Key idea: It balances minimizing prediction error with controlling the size of model coefficients.

Purpose in the project:
To determine whether regularization can improve the stability and predictive performance of the linear model.

---

3. Decision Tree Regressor

Decision Tree Regression is a non-linear supervised learning algorithm that predicts a continuous outcome by recursively splitting the data into smaller groups.

The algorithm selects suitable splitting rules at each stage to create groups with more similar target values. The process continues until a stopping condition is reached, and the final prediction is obtained at the leaf nodes.

Decision Trees can capture nonlinear relationships and interactions that traditional linear models may not capture effectively.

In your project, the Decision Tree was configured with a maximum depth of 10 to control model complexity.

Key idea: It learns a series of decision rules from the training data.

Purpose in the project:
To evaluate whether a non-linear tree-based approach can improve ticket price prediction compared with linear models.

---

4. Random Forest Regressor

Random Forest Regression is an ensemble machine learning algorithm that combines predictions from multiple Decision Trees.

Instead of depending on one tree, Random Forest constructs many trees using different samples and random subsets of information. The individual predictions are then combined, generally by averaging them for a regression problem.

This approach helps reduce the instability and overfitting associated with a single Decision Tree.

In your project, the Random Forest model used 100 trees with a maximum depth of 15.

Key idea: Multiple diverse Decision Trees are combined to produce a more stable and accurate prediction.

Purpose in the project:
To improve prediction accuracy and capture complex, nonlinear patterns in airline ticket pricing.

Project result:
Random Forest was the best-performing model, achieving $R^2 = 0.9765$, with MAE of 1,783 and RMSE of 3,476 on the test set.

---

5. Gradient Boosting Regressor

Gradient Boosting Regression is an ensemble learning algorithm that builds multiple Decision Trees sequentially.

Unlike Random Forest, where trees are largely developed independently, Gradient Boosting builds each new tree to improve the errors of the existing model. The contribution of each new tree is controlled using a learning rate.

This allows the algorithm to progressively improve its predictions and capture complex nonlinear patterns.

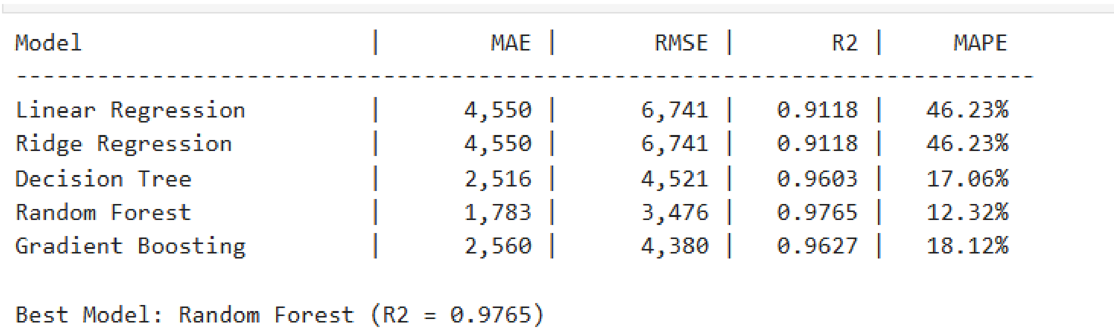
In your project, Gradient Boosting was configured with 100 estimators, a maximum tree depth of 5, and a learning rate of 0.1.

Key idea: Each successive tree attempts to correct the errors made by the previous ensemble.

Purpose in the project:
To evaluate a boosting-based ensemble method and compare its predictive capability with Decision Tree and Random Forest models.

Project result:
Gradient Boosting achieved $R^2 = 0.9627$, MAE of 2,560 and RMSE of 4,380 on the test set.

Model Comparison



Model	MAE	RMSE	R2	MAPE
Linear Regression	4,550	6,741	0.9118	46.23%
Ridge Regression	4,550	6,741	0.9118	46.23%
Decision Tree	2,516	4,521	0.9603	17.06%
Random Forest	1,783	3,476	0.9765	12.32%
Gradient Boosting	2,560	4,380	0.9627	18.12%

Best Model: Random Forest (R2 = 0.9765)

Interpretation:

Model Performance Interpretation

- Linear Regression: The model achieved an R^2 of 0.9118, with MAE of 4,550, RMSE of 6,741, and MAPE of 46.23%. It provides a reasonable baseline, but its relatively high prediction errors indicate that it is less effective for capturing the complex patterns in the dataset.
- Ridge Regression: Ridge Regression produced the same performance as Linear Regression, with an R^2 of 0.9118, MAE of 4,550, RMSE of 6,741, and MAPE of 46.23%. This indicates that the regularization applied in this model did not result in a measurable improvement over Linear Regression for this dataset.
- Decision Tree: The Decision Tree performed considerably better, achieving an R^2 of 0.9603, MAE of 2,516, RMSE of 4,521, and MAPE of 17.06%. The improvement indicates that the tree-based approach was able to capture nonlinear patterns more effectively than the linear models.
- Random Forest: Random Forest achieved the best overall performance, with an R^2 of 0.9765, MAE of 1,783, RMSE of 3,476, and MAPE of 12.32%. It has the

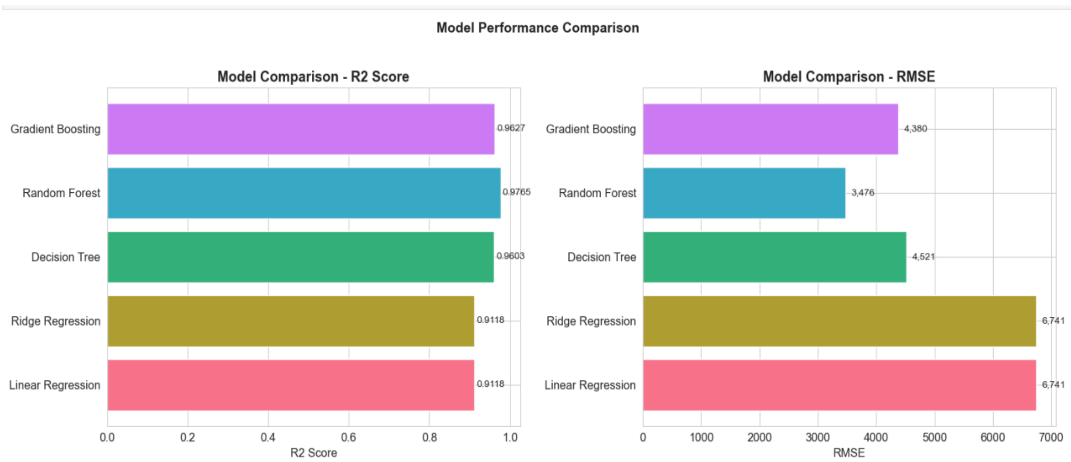
highest R^2 and the lowest error values among all the tested models, making it the most accurate model in this comparison.

- Gradient Boosting: Gradient Boosting also performed well, achieving an R^2 of 0.9627, MAE of 2,560, RMSE of 4,380, and MAPE of 18.12%. However, its performance was slightly lower than Random Forest and slightly better than or comparable to the Decision Tree depending on the metric.

Overall Interpretation

- The results show that the tree-based models performed better than the linear models.
- Random Forest is the best-performing model because it has the highest R^2 (0.9765) and the lowest MAE, RMSE, and MAPE.
- The results suggest that the underlying prediction task contains patterns that are better captured by nonlinear and ensemble-based models than by simple linear approaches.
- Therefore, Random Forest was selected as the final/best model for the prediction task.

Model Comparison Chart



Interpretation:

The chart shows that Random Forest performs the best, with the highest R^2 score (0.9765) and the lowest RMSE (3,476). Gradient Boosting and Decision Tree also show strong performance, while Linear Regression and Ridge Regression have lower R^2 and higher RMSE. Overall, the results indicate that Random Forest provides the most accurate and reliable predictions among the models compared.

Conclusion of objective 6

The five regression models were compared to determine the most suitable model for airline ticket price prediction. Linear Regression and Ridge Regression provided a baseline performance with an R^2 of 0.9118, while Decision Tree improved the performance with an R^2 of 0.9603. Gradient Boosting also performed well, achieving an R^2 of 0.9627. However, Random Forest performed the best, achieving the highest R^2 of 0.9765 and the lowest RMSE of 3,476, along with the lowest MAE and MAPE. Therefore, based on the model evaluation results in the provided Python notebook, Random Forest was selected as the best-performing model for airline ticket price prediction.

Conclusion

The study successfully analysed airline ticket pricing using EDA, statistical analysis, and predictive modelling. The results show that ticket prices are right-skewed and are influenced by factors such as airline, class, route, duration, stops, and days left before departure. The predictive analysis achieved strong performance, with the best approach obtaining an R^2 of 0.9765, RMSE of ₹3,476, MAE of ₹1,783, and MAPE of 12.32%. Overall, the study demonstrates that data-driven analysis can effectively explain pricing patterns and support accurate ticket-price estimation.

Recommendations

- Include seasonal trends, holidays, fuel prices, weather, and competition for more realistic analysis.
- Use larger and more diverse datasets to improve reliability and generalization.
- Apply hyperparameter tuning and cross-validation to improve prediction stability.
- Regularly update the analysis with recent pricing data because airline prices change dynamically.
- Develop a web or mobile application for practical real-time ticket-price estimation.

REFERENCE

- Koenker, R. (2005). *Quantile Regression*. Cambridge University Press. *(Used for quantile regression methodology)*
- McCullagh, P., & Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman & Hall. *(Used for Gamma regression model)*
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer. *(Used for regression, EDA, and ML concepts)*
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer. *(Used for advanced machine learning techniques)*
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. *(Used for Random Forest model)*
- Friedman, J. H. (2001). Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232. *(Used for boosting concepts in XGBoost)*
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *KDD Conference Proceedings*, 785–794. *(Used for XGBoost model)*
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. *(Used for implementation of ML models)*
- Seabold, S., & Perktold, J. (2010). Statsmodels: Statistical Modeling with Python. *Python in Science Conference*. *(Used for Quantile and GLM modeling)*
- Rohit Grewal. (2022). *Airline Flights Price Dataset*. Kaggle. https://www.kaggle.com/datasets/rohitgrewal/airlines-flights-data *(Dataset used for analysis and modeling)*

APPENDIX

Python code and Report file visit this link

Link: https://github.com/vishalbotadara4104/Semester_4_Projects

Scan QR Code For More Information

Tool Used: Python

MS Excel